Herded or Anchored? Testing Conformity in Machine Learning on a Decade of ICLR Submissions

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Abstract

Machine learning is often said to herd: researchers chase what is hot, and review favours it. But a field can converge without copying. Researchers may instead stay close to a record: their own earlier work, or a topic they already know. We separate these two routes to conformity, imitation and continuity, and test the herding account where they predict different things. The data are 43,480 ICLR submissions (2017–2026), rejected and withdrawn papers included, with hypotheses registered before outcomes were read and re-tested on 11,099 held-out papers. The field did converge on foundation models, but convergence fits both routes, and how many topics look emergent depends on hindsight: 8 to 29, depending on the years a topic map is fitted on. Where the routes part, no herding is detectable. Researchers new to a topic propose methods more often than incumbents do, the reverse of our registered prediction. At review, crowded topics show neither a detectable premium nor a penalty, while papers far from their authors' earlier work are accepted less often (post hoc, confirmed on held-out papers). The conformity we can detect at ICLR points to a record, not to the crowd.

1 Introduction

Machine learning is often said to herd. Between 2017 and 2026, yearly submissions to the International Conference on Learning Representations (ICLR) in our discovery corpus grew 40-fold, from 384 to 15,411. The share building on a foundation model (FM) rose from 0.3% to 54.4%, and the share proposing a new architecture fell from 24.5% to 7.5%. Keywords rose and fell (Figure 1): generative adversarial networks fell from 8.1% of submissions in 2018, and large language models reached 16.7% in 2026. The wave is NLP's too, as its researchers increasingly publish at ML venues (Jurgens, 2026). Whether science has fashions is an old question (Crane, 1969). Critics of ML scholarship point to incentives that favour short-term success (Lipton and Steinhardt, 2019) and to convergence on what suits the field's hardware, benchmarks and foundation models (Hooker, 2021; Dehghani et al., 2021; Bommasani et al., 2021); cascade and fashion theories explain waves by people taking their cue from others (Bikhchandani et al., 1992; Banerjee, 1992; Abrahamson, 1991).

Is the switch to FMs not herding enough? Convergence on a shared substrate is predicted by lock-in, by the efficient choice of a tool that works and by a bandwagon alike, so the curve cannot tell them apart. The herding question is about behaviour: what researchers write when they enter a topic, and what the review gate accepts. There, two routes to conformity part. *Imitation* takes its reference point from what others are doing now: what is popular, rising or recently accepted. *Continuity* takes it from a record: the team's own earlier work, or a topic that a team member already knows. Imitation makes teams alike; continuity keeps each team near its own past. We call work that conforms by continuity *anchored*.[1]

Read as herding, the waves make three claims: they are fashions; newcomers to a topic carry them by following others; and review favours following. We test each claim at its site, the field, entry to a topic and the acceptance gate, on 43,480 ICLR submissions from 2017–2026, rejected and withdrawn papers included. Hypotheses were registered before any full-corpus label existed, outcomes stayed sealed until one frozen run, and the main tests were run again on 11,099 held-out papers: a pre-registered out-of-sample replication, not a certified untouched holdout. Each finding carries one of three labels: REPLICATED (from a registered test,

[1] 'Anchored' names a position relative to a record, not a mechanism. It is not the anchoring-and-adjustment heuristic (Tversky and Kahneman, 1974), the sense in which writing on herding in science uses the word (Baddeley, 2015).

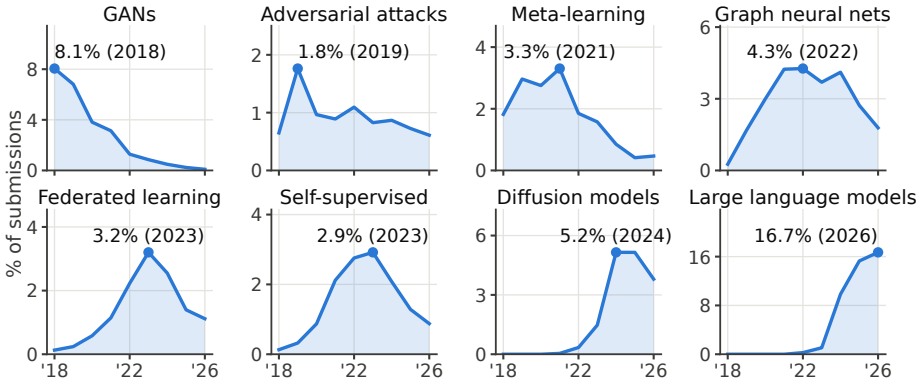


Figure 1: Research waves at ICLR: share of discovery submissions listing each author keyword, 2018–2026; dots mark the peak. The eight keywords were chosen for display after their curves were known. Across the 68 most frequent keywords, 17 peaked in 2019–2024 with shares of at most half the peak both before it and in 2026. Uses no acceptance data.

supported again on held-out papers), HELD-OUT SUPPORTED (found post hoc, then supported by a test declared before the held-out data were read) or EXPLORATORY (discovery only, or descriptive). Tests without a verdict are called inconclusive or uninformative (Section 3).

At the field level, keyword waves are real, but emergence depends on hindsight, and the durable shift, the FM package, fits every route. At entry, researchers new to a topic propose methods more often than incumbents, while, in the FM era, benchmark and evaluation work sits with incumbents: the opposite of the follow-on entry that herding implies, and of our own registered prediction. At the gate, no premium for crowded topics is detectable, the branded title's small premium lies below what the held-out test could detect, and what replicated, apart from regressing topic edges, is tied to the team's own record.

Prior work relates ICLR acceptance to prestige, process and research area (Tran et al., 2020; Ibrahim et al., 2026; Jung et al., 2025; Wang et al., 2023; Xu et al., 2026) and describes which papers start topics, who enters LLM research and how far researchers move from their own past work (Huang, 2026; Movva et al., 2024; Bao and Li, 2025). To our knowledge, among the works we checked, none tests the herding account at entry and at the gate within one submission population, with registered hypotheses re-tested on held-out papers, as Ibrahim et al. (2026) did for a prestige gap. We contribute:

- **Entry.** A registered test of herding where imitation predicts it most sharply. Researchers entering a topic propose methods more often (+3.7 pp; REPLICATED). The surplus holds for researchers with ICLR work in other topics and predates the FM wave, while, in the FM era, benchmark and evaluation work sits with topic incumbents and larger teams (EXPLORATORY).
- **Gate.** Using every decided submission under double-blind review, rejected ones included, acceptance goes with closeness to the team's own record among teams that have one (far jump -5.4 pp, held-out -6.1; HELD-OUT SUPPORTED), not detectably with the crowd. Hill et al. (2025) relate pivots to publication success; we add the accept/reject decision itself.
- **Method.** On the same papers, the number of emerging topics ranges from 8 to 29 depending on the years the topic map is fitted on (EXPLORATORY). Emergence is hard to judge before the fact (Rotolo et al., 2015; Li et al., 2025), and a prospective signal has been tested (Khanbayov and Kurban, 2026); an early-window refit is a cheap check on trend claims.

2 Two Routes to Conformity

Imitation. When outcomes are uncertain, organisations mimic one another (DiMaggio and Powell, 1983), individuals follow their predecessors and discard their own information (Bikhchandani et al., 1992; Banerjee, 1992), and herding protects a reputation when evaluators judge people against their peers (Scharfstein and Stein, 1990); bandwagons can spread even when adopting brings no gain (Abrahamson and Rosenkopf, 1993). In ML, benchmark use concentrates on a small set of datasets, increasingly so (Koch et al., 2021). Entrants to a topic face the most uncertainty, so imitation predicts that they flock to hot topics, adopt a rising package

Route (reference point)	Field	Entry	Gate
Imitation (others, now)	Convergence on a package	Entrants flock to hot topics, adopt the rising package early and write follow-on benchmark and evaluation work	A premium for crowded topics, the popular form and riding a wave
Continuity (a record)	Convergence on a package	Entrants bring other work; incumbents consolidate; the package travels with a record	Acceptance tracks closeness to the team's own record

Table 1: What each route predicts at the three sites. Lock-in and efficient choice also predict convergence on a package, so only entry and the gate can tell the routes apart. Imitation's entry and gate predictions were registered before outcomes were read; continuity's were written after the tests had run.

early, and write follow-on work that benchmarks or evaluates what others proposed. At the gate, if review favours following, papers in crowded topics, in the popular form or riding a wave are accepted more often.

Continuity. Search stays near existing competence (March, 1991; Stuart and Podolny, 1996), and taking up new knowledge depends on related prior knowledge (Cohen and Levinthal, 1990). At entry, continuity predicts that incumbents consolidate a topic's record and that a new package travels with a record; on outsider accounts, entrants bring work from elsewhere (Zeng et al., 2021; Packalen and Bhattacharya, 2019). At the gate, acceptance should track how close a paper stays to its team's record. Weisberg and Muldoon (2009) separate 'followers' of others' approaches from agents who ignore them; we borrow the distinction, not their welfare conclusions (Alexander et al., 2015).

Rivals, and when the predictions were written. Visible status, cumulative advantage and expertise (Merton, 1968; Ibrahim et al., 2026; Li, 2017) also predict that acceptance goes with a team's record. Network diffusion, in which less connected actors adopt later (Coleman et al., 1957), also predicts that teams new to ICLR take up a package later, so that contrast cannot separate the routes. Imitation's entry and gate predictions (Table 1) are the herding hypotheses we registered before reading any outcome; continuity's were written after the tests had run. The distinction organises the tests; it did not generate them. To our knowledge, among the works we checked, the two routes have not been tested against each other at entry or at the gate.

3 Data and Design

Corpus. The corpus comes from our own dated acquisitions of official OpenReview records (Appendix A): 43,480 discovery submissions with a usable abstract and 11,099 held-out papers. Acceptance tests use the 32,190 decided discovery papers from 2018–2026, 40.1% of them accepted, and 8,185 decided held-out papers from 2018–2026; withdrawn papers are left out.

Instruments. A *topic* is one of 100 spherical k-means clusters of SPECTER2 embeddings of title and abstract (Cohan et al., 2020; Singh et al., 2023), fitted on all years unless another map is named. An LLM with a fixed codebook gives each paper one primary contribution type: a *method paper* proposes a method, including a new architecture, and a *measurement paper* contributes a dataset, benchmark or evaluation; an *FM paper* uses or adapts a large pre-trained model. A blind second pass agreed at κ 0.93, and the author hand-checked a small random sample. This is consistency between LLM labellings, not human-level validity; zero-shot LLM labels reach only fair agreement with human coders on many tasks (Ziems et al., 2024).

Entry profiles. Entry status is relative to a topic and counts ICLR submissions only. In a *newcomer paper*, no author has an earlier ICLR submission, with any outcome, in the paper's topic (a topic-relative definition, as in Biliotti et al., 2024); otherwise it is an *incumbent paper*, and an author with such a submission is a *topic insider*. A newcomer paper comes from a *first-time team* if no author has any earlier ICLR submission, and from a *returning topic newcomer* if some author has one in another topic.

Two distances. Both distances use one centred SPECTER2 space, with different reference points. *Closeness to the team's own record*, $F_{\max}$, is the highest cosine between the paper and the mean of an author's earlier ICLR submissions since 2021, taken over the paper's *experienced authors*, those with such submissions (following Bao and Li, 2025). *Distance from earlier ICLR work* is one minus the mean cosine to the ten nearest submissions of earlier years, as a percentile within topic and year.

Prospective measurement. Predictors use only the past: *prior FM adoption* is the FM share of the topic's previous-year papers, *crowding* is the

log of one plus the topic's earlier papers, and a *hot topic* is in the top fifth of prior growth. Code links leak the outcome into 17.6% of accepted but 0.5% of rejected 2023–2026 abstracts (Kaufman et al., 2012); in discovery, the distance-based gate estimates hold without them (no link-removed text exists for held-out papers; Appendix F).

Protocol. Hypotheses, models and a support rule were registered before any full-corpus label existed. A hypothesis is supported if it has the predicted sign, a Holm-adjusted $p < 0.05$ in its family (Holm, 1979), and that sign in at least 75% of counted variants and in both hindsight variants (a year-balanced map; the 2017–2020 map applied to 2021–2026 papers). Outcomes could be read only by committed code in one logged, gated run (Nosek et al., 2018; van den Akker et al., 2021). Post hoc analyses were declared before they were run, and their one-sided held-out tests before any held-out estimate for that question was computed. Models are linear probability models with topic and year fixed effects and topic-clustered errors. We disclose two weaknesses. Under a simulated null whose structure cuts across topics, the main entry tests rejected 12.6% and 10.0% of the time at a nominal 5%, which a wild cluster bootstrap (Cameron et al., 2008) did not fix (Appendix H). And keyword popularity and title style had been explored against acceptance on the discovery papers before registration: the discovery estimate for titles is not a test, and the keyword-timing tests probe a dimension already seen. A test that is not supported and whose estimate lies below its minimum detectable effect at 80% power (MDE80) is *uninformative*: it shows neither an association nor its absence. Supported held-out tests confirm a direction, not a size.

4 The Field: What Is Fashion and What Lasts

Keyword waves are real, but how many topics look emergent depends on hindsight, and the one durable shift, the FM package, fits every route.

Waves. Author keywords need no topic map, and at their level the waves are real: of the 68 most frequent keywords, 17 rose and fell, peaking in 2019–2024 with shares of at most half the peak both before it and in 2026 (Figure 1; EXPLORATORY).

Emergence depends on hindsight. Histories of the field usually fit one topic model to the whole

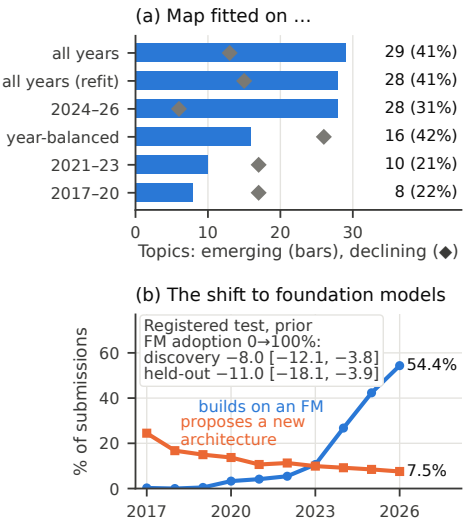


Figure 2: (a) Emerging (bars) and declining (grey diamonds) topics on $k = 100$ maps fitted on different years, same 43,480 papers; right-hand labels: emerging count (share of 2024–2026 papers in emerging topics, 22.3% on the 2017–2020 map against 40.9% on the all-years map). All years (refit): the all-years map refitted with another seed. Topic size 'predicts' two-year growth only on the all-years map (R^2 0.28 against −0.03; no paired test). (b) Share of submissions building on an FM (blue circles) and proposing a new architecture (orange squares); LLM labels. Inset: the registered test of the package shift, REPLICATED (held-out confirms direction, not size). Uses no acceptance data.

archive (Hall et al., 2008; Anderson et al., 2012; Klinger et al., 2020), and dynamic topic models and BERTopic fit all years at once (Blei and Lafferty, 2006; Grootendorst, 2022). We call a topic *emerging* if its 2024–2026 share is at least three times its 2017–2020 share. On the same papers and pipeline, a map fitted on 2017–2020 finds 8 emerging topics and a map fitted on all years finds 29 (Figure 2a, Appendix C; EXPLORATORY). Two reasons combine, and we cannot separate them. A map fitted on all years leaks the future into the past, like temporal leakage in prediction (Kapoor and Narayanan, 2023) and hindsight in judgement (Fischhoff, 1975): with 69% of papers from 2024–2026, it is in effect a present-day map. And the early map cannot represent topics born after 2020. The two counts therefore bracket the number of emerging topics rather than fix it.

What lasts: a package. The FM share rose every year from 2018 (Figure 2b). Where more of a topic's previous-year papers built on FMs, fewer of its papers proposed new architectures: −7.98 pp [−12.14, −3.82] for prior adoption from 0 to 100%

in discovery and −11.0 [−18.1, −3.9] on held-out papers (REPLICATED; the held-out test confirms the direction, not the size). No detectable shift towards empirical or evaluation papers came with it (−0.63 [−3.64, +2.38]; not supported, inconclusive), and the association may also run the other way. The pattern matches accounts of NLP's paradigm shifts (Gururaja et al., 2023; Saphra et al., 2024; Pramanick et al., 2023). But lock-in (Hooker, 2021), efficient choice (Sutton, 2019) and a bandwagon (Fujimura, 1988) all predict such convergence, so the field level cannot tell imitation from continuity. Entry and the gate must.

5 Entry: Who Writes What

Entry shows a division of labour: researchers entering a topic bring methods, while, in the FM era, benchmark and evaluation work sits with larger teams and topic incumbents. This is the opposite of the follow-on entry that the herding account implies.

Newcomers propose methods. We registered that newcomer papers, whose authors face the most uncertainty, would more often be follow-on work: benchmarks, datasets and evaluations of what others had proposed. Newcomer papers are 72.9% of the 42,325 papers from 2019–2026 with an author identifier. They are more often method papers, +3.71 pp [+2.61, +4.80] in discovery and +3.60 [+1.48, +5.71] held-out, and less often measurement papers, −1.70 [−2.46, −0.93] and −2.10 [−3.35, −0.85] (Table 2, Figure 3; REPLICATED). Our registered prediction thus failed in the opposite direction, and the reversed hypotheses, registered before any held-out data existed, held on held-out papers. All 10 counted variants agree, and the t-statistics (6.7 and −4.4) are large against the inflated test size (Appendix H). Much of the measurement gap is accounted for by team size: with log team size it shrinks to −0.54 [−1.11, +0.04] in discovery and −0.98 held-out, while the method gap does not move (+3.92 and +3.64).

Entering a topic, not ICLR. Newcomer papers split into first-time teams (26.3% of papers; median team of 3) and returning topic newcomers (46.7%; median 5); incumbent papers are the other 27.0% (median 5). Returning topic newcomers write method papers more often than incumbents, +3.43 pp [+2.31, +4.55], at the same median team size and also with team size controlled (+3.62).

Contrast	Discovery [95% CI]	Held-out [95% CI]	Label
Method paper	+3.71 [+2.61, +4.80]	+3.60 [+1.48, +5.71]	REPLICATED
Measurement paper	−1.70 [−2.46, −0.93]	−2.10 [−3.35, −0.85]	REPLICATED
Measurement paper, with log team size[a]	−0.54 [−1.11, +0.04]	−0.98	registered variant
Method: returning topic newcomer − incumbent[b]	+3.43 [+2.31, +4.55]	–	EXPLORATORY
Method: first-time team − returning topic newcomer[b]	+0.92 [−0.44, +2.29]	–	EXPLORATORY
Distance from earlier ICLR work, own work excluded (percentile points)	+4.33 [+3.44, +5.22]	–	EXPLORATORY
Flocking: incumbent share, hot − other topics	−0.73 [−2.68, +1.22]	not registered	inconclusive

Table 2: Entry: newcomer minus incumbent papers unless the row names other profiles; pp within topic and year. Held-out tests are one-sided and confirm direction, not size. [a]Registered supplementary variant, outside the support rule. [b]First-time team: no author has any earlier ICLR submission; returning topic newcomer: an author has one, in another topic (Appendix E).

First-time teams do so by +4.35 [+2.82, +5.89], and the two newcomer profiles do not differ detectably (+0.92 [−0.44, +2.29]; all EXPLORATORY). The method surplus thus goes mainly with entering a topic, not with being new to ICLR or with small teams, and it is present in every era, so it predates the FM wave (Appendix E). The measurement gap is different. For first-time teams it is accounted for by team size (−2.24 [−3.24, −1.25] against incumbents, +0.09 [−0.63, +0.80] with team size); for returning topic newcomers part of it remains (−1.47, and −0.81 [−1.46, −0.16] with team size). It appears only in 2024–2026 (first-time teams against incumbents −2.75 [−4.07, −1.44]). In the FM era, then, benchmark and evaluation work sits with topic incumbents and larger teams; team size is the only resource we measure. NLP shows a parallel at the level of research trends, where tasks and methods lead and datasets follow (Pramanick et al., 2023).

Farther from earlier work. Newcomer papers also sit farther from earlier ICLR work: +5.90 percentile points [+4.95, +6.85] within topic and year, +5.24 among method papers alone, and +4.33 [+3.44, +5.22] with the authors' own earlier papers excluded from the reference set (EXPLORATORY). Papers by topic newcomers with an ICLR record are also less likely than incumbent papers to have a same-year *twin*, an independent submis-

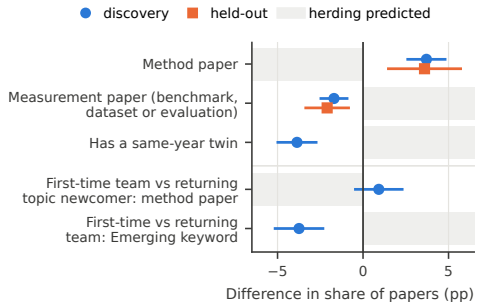


Figure 3: Entry contrasts within topic and year (pp) with 95% intervals: discovery (blue circles) and held-out (orange squares). Rows 1–2, newcomer minus incumbent papers, are registered and REPLICATED (held-out confirms direction, not size). Rows 3–5 are EXPLORATORY (discovery only): row 3 compares papers by topic newcomers with an ICLR record with incumbent papers (first-time teams controlled separately; class, FM and team-size controls); row 4 compares first-time teams with returning topic newcomers, and row 5 with all teams that have an ICLR record. Grey: the side the herding prediction expected. Uses no acceptance data.

sion at cosine ≥ 0.70 (with first-time teams controlled separately): −3.86 pp [−4.98, −2.74] (EXPLORATORY). The follow-on prediction expects newcomer papers to sit closer to earlier work, not farther. Writing method papers could itself imitate the venue's dominant genre, three quarters of ICLR papers, but newcomers' distance from earlier work tells against that reading. A method paper is a contribution type, and embedding distance is not originality: these measures give no sign that newcomers follow, and do not show that they are more original.

Later onto the package. In 2023–2026 topics at 5–50% prior FM adoption, first-time teams write FM papers less often than returning topic newcomers, by −4.58 pp [−6.55, −2.60] up to −14.71 [−19.82, −9.61] as adoption rises, while returning topic newcomers do not differ detectably from incumbents (Figure 5 in Appendix E). First-time teams also list young, rising *Emerging* keywords less often than teams with an ICLR record: −3.74 [−5.14, −2.35] (both EXPLORATORY). The package travels with an ICLR record, as absorptive capacity predicts (Cohen and Levinthal, 1990); network diffusion predicts the same order, so this contrast does not discriminate.

What stays open. Whether entrants crowd into hot topics is unsettled: the registered flocking test, the incumbent share in hot against other topics, is inconclusive (−0.73 [−2.68, +1.22]). At entry, no detectable difference fits the follow-on prediction.

6 The Gate: Crowd or Record?

At the gate we detect no premium for following the crowd. Apart from topic edges, which regress, what replicated is tied to the team's own record, which continuity predicts and status and expertise predict too.

Does acceptance go with following the crowd? The registered crowding test predicted a penalty, and it is uninformative: −1.68 pp per log unit (natural log of one plus the topic's earlier papers) [−3.68, +0.32] in discovery and −1.51 [−5.20, +2.19] held-out, against an MDE80 of 4.8 (Table 3, Figure 4). Neither a penalty nor a premium is detectable, and held-out premiums above +2.19 pp per log unit lie outside the interval.

The plainest imitation-like spread in the corpus is in form. The share of titles that open with a short coined name and a colon rose from 10.9% in 2017 to 44.6% in 2026 (Appendix G). Within topic and year, with class, FM and team-size controls, such branded titles are accepted +1.74 pp [+0.37, +3.11] more often in discovery, which is not a test, since title style had been explored before registration. The declared held-out test gives +1.76 [−0.82, +4.34] (one-sided p 0.089, MDE80 3.2): not supported, and uninformative, so the premium stays EXPLORATORY. Raw within-year gaps are larger (+3.3 in discovery, +4.0 held-out); about half goes with topic, type, FM use and team size. Titles can also be revised after the decision, which could create such a premium.

FM papers are accepted more often, +4.62 pp [+2.79, +6.44], with no detectable difference between late and early adopters: the premium changes by +0.19 [−0.96, +1.34] per logit unit (log-odds of the topic's prior FM adoption; EXPLORATORY), which fits rewarded imitation and efficient choice alike. A topic's *edge*, its acceptance minus that of all other papers of the same year (topics of the 2017–2020 map), is real but regresses: the 2018–2022 edge predicts the 2024–2026 edge with slope 0.38 [0.18, 0.58] among discovery papers, and the held-out papers' 2024–2026 edges regress on the discovery 2018–2022 edges with slope 0.60 [0.32, 0.88] (HELD-OUT SUPPORTED; two different comparisons). A favoured topic keeps part of its edge, not all of it, as expected when decisions carry a large arbitrary component (Cortes and Lawrence,

Acceptance and …	Discovery [95% CI]	Held-out [95% CI]	Label
Others: the crowd, the form, the package			
Crowding, per log unit[a]	−1.68 [−3.68, +0.32]	−1.51 [−5.20, +2.19]	uninformative
Branded title[b]	+1.74 [+0.37, +3.11]	+1.76 [−0.82, +4.34]	EXPLORATORY
FM premium, per logit unit of prior adoption	+0.19 [−0.96, +1.34]	–	EXPLORATORY
Topic edge 2018–22 → 2024–26, slope[c]	0.38 [0.18, 0.58]	0.60 [0.32, 0.88]	HELD-OUT SUPPORTED
Emerging − Mainstream keyword[b,c]	+2.67 [+0.76, +4.58]	+3.25 [−0.18, +6.69]	EXPLORATORY
The team's own record			
Far jump: $F_{\max}$ p10 − median	−5.44 [−7.10, −3.78]	−6.09 [−9.32, −2.86]	HELD-OUT SUPPORTED
Same author, +0.1 cosine closer[c]	+0.55 [+0.21, +0.90]	+1.06 [+0.23, +1.89]	HELD-OUT SUPPORTED
One topic insider vs none[c]	+3.42 [+1.18, +5.66]	+3.88 [+0.28, +7.47]	HELD-OUT SUPPORTED
The field's record			
Least-distant fifth − middle fifth	−3.27 [−4.93, −1.60]	−0.13 [−3.50, +3.25]	uninformative

Table 3: The gate: pp within topic and year (topic edges and keywords: against the same year's other papers). Rows differ in population and unit: compare replication status, not size. Held-out tests are one-sided and confirm direction, not size. [a]Registered; predicted a penalty; MDE80 4.8. [b]Explored against acceptance before registration. Titles: the discovery estimate is not a test; held-out MDE80 3.2. Keywords: the discovery estimate is a declared post hoc test on a dimension already seen. [c]Holm across the six secondary held-out tests: topic edge 0.019, same author 0.026, insider 0.052, Emerging keywords 0.063.

2021; Beygelzimer et al., 2023). Keyword timing is weak (EXPLORATORY; Appendix D): on held-out papers, *Emerging* keywords go with higher acceptance than old, flat *Mainstream* ones (+3.25 [−0.18, +6.69]), and acceptance may fall after a keyword's share peaks (−5.2 [−11.1, +0.7]; −1.7 without the six keywords inspected first). Neither shows the flat premium that rewarded imitation of what is popular predicts.

What does acceptance go with? The clearest pattern is the *far jump*. Among 20,430 decided 2023–2026 papers with an experienced author, acceptance at the 10th percentile of closeness to the team's own record minus that at the median is −5.44 pp [−7.10, −3.78] in discovery and −6.09 [−9.32, −2.86] held-out, with power 0.89 (HELD-OUT SUPPORTED; the declared primary test). Adjusted acceptance rises with closeness up to the fourth fifth (Figure 6 in Appendix F). The far jump holds with a newcomer control (−5.0), although closeness correlates −0.57 with the newcomer flag. Holding the author fixed, a paper 0.1 closer in cosine to that author's earlier work goes with acceptance +0.55 pp [+0.21, +0.90] higher in discovery and +1.06 [+0.23, +1.89] held-out (HELD-OUT SUPPORTED; it survives a family-wise correction across the secondary tests). Controls for newcomer status and for co-authors' experience move it little (+0.48 and +0.50), though a confounder in co-author quality could also bias it: in a simulation reported in its declaration, such a confounder moved the estimate. Likewise, the same author is accepted about 1.4 pp less often [−2.7, −0.0] on a paper in a topic new to them (author fixed effects; EXPLORATORY).[2]

One topic insider on the team, against none, goes with +3.42 pp [+1.18, +5.66] in discovery and +3.88 [+0.28, +7.47] held-out, at fixed team size and number of experienced authors (HELD-OUT SUPPORTED). It does not survive the family-wise correction (Holm 0.052), and a status reading fits it as well: per head, an insider goes with only +0.58 pp more than an experienced author from another topic (p 0.16; EXPLORATORY).

The field's record is undetermined (uninformative). In the same space, the least-distant fifth of papers against the middle fifth is −3.27 [−4.93, −1.60] in discovery but −0.13 [−3.50, +3.25] held-out, with power 0.47 (uninformative), and the within-author slope for distance from the field was not supported either. Of the distance-based tests, only the own-record associations replicated.

Readings. These associations are referenced to a record, as continuity predicts, but expertise, evaluator conservatism and visible status predict them as well (Li, 2017; Merton, 1968). Double-blind review does not rule status out: where reviewers see the authors, famous authors fare better (Tomkins et al., 2017), and preprints can reveal who wrote a paper (Bharadhwaj et al., 2020). Pivots have been related to publication success (Hill et al., 2025); we observe the accept/reject decision itself, for every decided submission.

7 Discussion: Where Conformity Lives

What the evidence supports. The registered herding predictions failed at entry (follow-on work) or stayed inconclusive (flocking), and the registered crowding test at the gate was uninformative. Every association that replicated at the gate is referenced to the team's own record, except persisting topic

[2]Newcomer papers as a whole are accepted 10.2 pp less often [−11.8, −8.7] within topic and year (EXPLORATORY), but that gap mostly reflects who writes them (Appendix F).

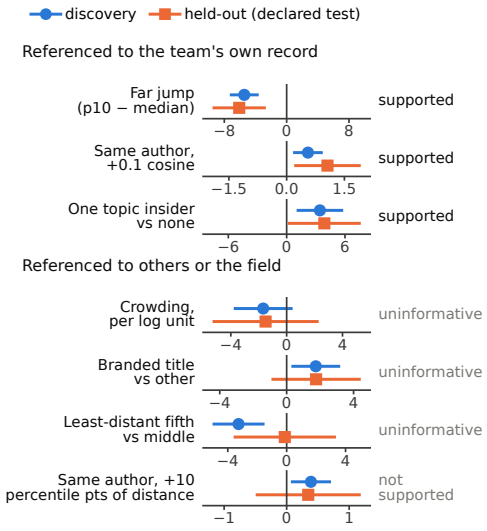


Figure 4: Gate tests by reference point: discovery estimates (blue circles) and held-out declared tests (orange squares), 95% intervals; the held-out verdict is at right (supported: HELD-OUT SUPPORTED; uninformative: not supported and below the test's MDE80; not supported: a declared test without a registered MDE80). Each row has its own scale: pp for the far jump, insider, branded title and least-distant fifth; pp per log unit for crowding; per 0.1 cosine and per 10 percentile points for the two same-author slopes. Compare replication status, not size. Crowding was registered; title style had been explored before registration; the other rows were found post hoc in discovery. Each held-out test was declared before any held-out estimate for that question was computed.

edges, which fit imitation of past acceptance and a rival (regression to the mean in noisy review) alike (Table 4). The conformity we do detect is a pull towards a record rather than towards identical work: incumbents' papers sit near their own earlier work, and acceptance goes with closeness to the team's record. But the record-side gate results do not single out continuity. Status and expertise predict them as well, and the test that could separate them, whether they concentrate where a preprint identifies the authors (Ibrahim et al., 2026), has not been run.

What remains open. Whether entrants crowd into hot topics is unsettled, and the absence of a detectable premium does not rule out bandwagons that bring no premium (Abrahamson and Rosenkopf, 1993); the spread of branded titles may be one, or it may carry a small premium; the held-out test cannot tell. Acceptance is also a short-run outcome: in the previous shift, researchers who stayed close to their own record had lower later impact (Bao and Li, 2025), so acceptance and impact need not go with the same work (cf. March, 1991; Rzhetsky et al., 2015).

	Entry (what is written)	Gate (acceptance)
Field, both routes: convergence on the FM package (REPLICATED), also predicted by lock-in and efficient choice.		
Imitation (others, now)	Follow-on prediction reversed (REPLICATED); newcomers farther from earlier work; fewer twins for returning topic newcomers (EXPLORATORY); flocking inconclusive.	No detectable crowding premium (uninformative); branded titles: small premium, held-out uninformative (EXPLORATORY). Fit a rival too: edges partly kept (HELD-OUT SUPPORTED); no detectable change in the FM premium; timing weak (EXPLORATORY).
Continuity (a record)	Incumbents write more measurement papers (REPLICATED; much of it accounted for by team size); the package travels with an ICLR record, as diffusion also predicts (EXPLORATORY).	Lower far from the team's record; higher for the same author's closer work and with a topic insider (HELD-OUT SUPPORTED; status and expertise fit too).

Table 4: Where conformity shows, by route and site: a summary of Sections 4–6, with no new estimates. The field's record at the gate is uninformative.

Why the distinction matters. The two routes carry different risks. Imitation can discard private information and leave a fragile consensus (Bikhchandani et al., 1992; Banerjee, 1992); continuity trades long-run exploration for short-run competence (March, 1991). A field that looks herded may conform by continuity instead, and a diagnosis of fads would then misread it. For histories of NLP and ML (Hall et al., 2008; Anderson et al., 2012), an early-window refit and lagged predictors are a cheap way to see which trend claims survive hindsight.

8 Conclusion

We tested the herding account on a decade of ICLR submissions. At the field level, keywords rose and fell, emergence depends on hindsight, and the durable shift, the FM package, fits every route. At entry, researchers new to a topic bring methods, and, in the FM era, benchmark and evaluation work sits with incumbents and larger teams. At the gate, no premium for crowded topics is detectable, and acceptance goes with closeness to the team's own record. The conformity we can detect at ICLR points to a record, not to the crowd.

Limitations

All results are associations at one venue; none measures what choosing a topic, team, title or contri-

bution type would change. The held-out set shares conference-years, reviewer pools and authors with the discovery papers, so it guards against analyst overfitting but not against year- or reviewer-specific patterns, and supported tests confirm direction, not size. The imitation–continuity frame was drawn up after the registered and declared tests had run; continuity's predictions were written afterwards. The record-side gate results were found post hoc, two of the three are secondary held-out tests, and the insider association does not survive a family-wise correction. Under one simulated null the main entry tests rejected 12.6% and 10.0% of the time at a nominal 5%. Contribution and FM labels are LLM labels, consistent across passes and hand-checked by the author on a small sample only; a method paper is a contribution type, not a measure of novelty, an adapter on an LLM counts as a method, and embedding distance is not originality. Entry status counts ICLR submissions only, and the 2021 switch from e-mail to profile identifiers hides earlier histories, which pulls entry contrasts towards zero. Abstracts and titles are partly camera-ready: links were removed and the text re-embedded for discovery papers only, and titles could not be corrected. Withdrawn papers are left out of acceptance tests. Status and continuity are not separated: work that continues a known line may itself reveal its authors, and we did not run the test of whether the record-side associations concentrate where a preprint identifies the authors. The hindsight comparison is a bracket, because the maps differ in fitting corpus as well as window. Keyword popularity and title style had been explored against acceptance before registration.

Ethics Statement

Author identifiers were used only as keys to compute entry status, closeness and the exclusion of a paper's own authors' earlier work; no author is named and all results are aggregates. The data are public OpenReview records, collected in our own dated acquisitions under the platform's terms. No review scores are used. Contribution labels were produced by an LLM (Claude Haiku) with a fixed codebook; agreement figures compare LLM labellings, not human judgement.

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A Data Provenance

The herding account is tested on one complete submission population, so that neither the entry contrasts nor the gate tests condition on acceptance. This appendix documents where the papers come from and which of them each test uses.

Acquisition. The corpus comes from our own dated acquisitions of official OpenReview records, reconciled into 55,938 submission instances. Titles, abstracts, author-declared keywords, primary areas, author lists and decisions all come from the selected official submission metadata; the keyword preprocessing is our own. The discovery export holds 43,516 submissions and 219,719 declared author-list positions (positions, not distinct people). The discovery corpus is the 43,480 submissions with a usable abstract of at least 30 words; 34 placeholders and 2 malformed abstracts are dropped. Abstracts are a September 2026 snapshot, so accepted papers more often carry camera-ready text: in 2023–2026, 17.6% of accepted and 0.5% of rejected abstracts contain a code link (Appendix F reports the link-removed re-run).

Splits. Table 5 gives papers by year. The held-out split is the 11,113 released former-holdout papers, of which 11,099 have an abstract, released on 2026-10-18. They were this study's confirmation split and had not been analysed before the held-out tests were registered. The source policy does not certify an untouched, independent holdout. The source does not document how papers were assigned to the split, and it excluded 1,309 pending family or link cases. We therefore call the held-out tests a pre-registered out-of-sample replication, not a certified untouched holdout. The two splits share conference-years, topic-year cells, reviewer pools and authors, so the held-out tests guard against analyst overfitting but not against year- or reviewer-specific patterns. The pooled corpus (54,579 papers) is used only for the exploratory re-run (Appendix D).

Analysis populations. Field analyses use all discovery papers (the registered field tests: 43,096 papers from 2018–2026; H1d and H1e 42,990). Entry analyses use the 42,325 papers from 2019–2026 with at least one author identifier (42,315 in 776 topic-year cells after singleton cells are dropped). Acceptance tests use the 32,190 ordinary submissions from 2018–2026 with a final accept or reject decision (40.1% accepted); the 9,986 withdrawn submissions are left out. Held-out populations are 10,978 papers (H1d), 10,718 (entry) and 8,185 decided held-out papers from 2018–2026 (acceptance; 8,284 decided in all). The post hoc gate modules use decided 2023–2026 papers (Appendix F).

Year	Discovery	Held-out
2017	384	105
2018	770	212
2019	1,248	309
2020	2,070	500
2021	2,360	603
2022	2,649	704
2023	3,872	995
2024	5,765	1,445
2025	8,951	2,284
2026	15,411	3,942
All	43,480	11,099

Table 5: Papers with a usable abstract by year. Discovery: 2017–2020 4,472; 2021–2023 8,881; 2024–2026 30,127 (69.3%). Submissions grew 40.1-fold from 2017 to 2026. Outcome-free.

Existing collections, audited and not pooled. Re^2 (Zhang et al., 2025) was initially considered as the main dataset and audited: its main-conference ICLR coverage ends in 2023 and omits 2019, its conversations contain training-format placeholders, and its rebuttal train and test partitions share 475 paper identifiers. The audit motivated our own collection. A public GitHub archive of ICLR rebuttal records (`iclr-rebuttal-analysis`)[3] served as a separate layer for source comparison and identity reconciliation; it added no submissions and supplies no text. A public metadata archive (Paper Copilot)[4] was used for metadata comparisons and documents the lineage of reconstructed 2024–2025 review-score pairs; this study uses no review scores. None of the three is pooled into the analytical corpus.

What this means for the test. Because rejected and withdrawn submissions are present, what topic newcomers write is observed for every paper they submitted, and the gate tests compare accepted with rejected papers of the same topic and year. Neither half of the herding account is tested on survivors only.

[3] https://github.com/MlouisBE/iclr-rebuttal-analysis

[4] Described in arXiv:2510.13201.

k	Track.	NMI	ARI	Syn.	Coh.	NMI_{100}
50	100%	0.294	0.77	0.77	0.26	0.72
75	97%	0.302	0.74	0.72	0.27	0.80
100	**94%**	**0.302**	**0.68**	**0.67**	**0.29**	1
150	76%	0.301	0.62	0.67	0.31	0.78
200	53%	0.302	0.57	0.64	0.33	0.75

Table 6: Topic quality by k (sensitivity sweep). Track.: papers in trackable topics; NMI with ICLR primary areas; ARI across seeds in the sweep (the main map's own seed ARI is 0.66); Syn.: synonym overlap; Coh.: keyword coherence; NMI_{100}: NMI with the $k = 100$ map. Outcome-free (E).

Test (unit)	$k = 75$	$k = 100$	$k = 150$
H1d (pp, 0→100% FM)	−8.70	−7.98	−7.47
H2b (pp)	+3.06	+3.71	+3.87
H2c (pp)	−1.51	−1.70	−1.81
H2a (pp)	−1.39	−0.73	+0.37
H3a (pp per log unit)	−1.54	−1.68	−1.33

Table 7: Registered discovery estimates at three values of k (counted variants). Intervals for $k = 100$ are in Table 13. H3a is inconclusive at every k.

B Measurement Validation

Each prediction that separates imitation from continuity is measured with four instruments: topics, contribution labels, entry status, and distances in one embedding space. This appendix reports how far each can be trusted.

Topics. Title and abstract ("title [SEP] abstract") are embedded with SPECTER2 (Cohan et al., 2020; Singh et al., 2023) (base model with the proximity adapter), reduced by unit-length PCA to 128 dimensions (93% of the variance), and clustered by spherical k-means with $k = 100$ (best of 3 seeds $\times$ 20 initialisations). A broad level (20 topics, Ward) and a fine level (198 topics) are supplementary. The topic layer reads no acceptance, decision or review field; an output validator checks this. Topics hold 127–1,121 papers; 91 topics (94% of papers) have at least 30 papers in at least three years. On 2024–2026 papers, normalised mutual information (NMI) with ICLR's primary areas is 0.302, against 0.272 for a TF-IDF/LSA word-count baseline, and synonym overlap is 0.67 against 0.46. In a blind name check, the correct topic was picked for 80% of items at this level and for 61% at the fine level (chance 10%). The adjusted Rand index (ARI) across seeds is 0.66 (0.63–0.69); the median Jaccard overlap is 0.66, with 26 topics below 0.5. Table 6 gives the sensitivity to k: $k = 100$ was fixed before the sweep (a design choice, not tuned), and adjacent solutions nest only partly (76–82% of a finer topic's papers fall in one coarser topic). Table 7 shows that the registered estimates keep their sign at $k = 75$ and $k = 150$, except the inconclusive H2a at $k = 150$.

Contribution labels. Each paper receives one primary contribution type from a fixed LLM codebook (prompt v1): method (including new architectures, type "model"), theory, empirical, dataset/benchmark, evaluation, other. A measurement paper is a dataset/benchmark or evaluation paper. An FM paper uses or adapts a large pre-trained model (an LLM, CLIP or a pre-trained diffusion model), coded with the same codebook. Labels were produced by Claude Haiku subagents with the prompt unchanged, in batches of 100; this replaced a planned external run and is a logged deviation. The registered quality gate required at least 93% agreement with a 300-paper pilot and a second-pass $\kappa \geq 0.80$; the run achieved 97.7% and κ 0.93. Against a reference set of 2,300 papers labelled by LLM annotators, main-class agreement is 95.8% (pilot 300: 97.7%; 200 double-labelled training papers: 95.5%; 1,800 single-labelled: 95.6%). A blind second pass on 2,174 discovery papers gives κ 0.93 overall, 0.933 for method (97.5% agreement) and 0.949 for measurement (99.3%). Relative to the LLM reference on its random stratum, false-positive and false-negative rates are 0.084 and 0.011 for method, and 0.005 and 0.070 for measurement; the S8 variant rescales the entry estimates by them, which can only enlarge them (+4.09 and −1.84 pp). On 555 held-out papers, agreement is 96.9% for the main class (κ 0.931), 98.0% for method (0.949), 99.1% for measurement (0.923) and 98.2% for FM (0.960). These figures show consistency between LLM labellings more than human-level validity. The author also hand-checked a small random sample of labels against the codebook. No agreement figure between the LLM and a human annotator is available. Over all discovery papers, the class mix is method 75.1%, empirical 6.9%, theory 6.9%, dataset/benchmark 5.2%, other 3.5% and evaluation 2.4%; the measurement share rose from 2.6% (2017) to 11.1% (2026) (E).

Keywords. Author keywords are canonicalised to 40,930 expressions. The 912 keywords with at least 20 papers were typed by two LLM annotators

(κ 0.93; 96.6% agreement; 31 adjudicated). Keyword states use a 759-keyword vocabulary. Held-out papers carry raw keywords only; a lookup learned on discovery papers alone maps 45,778 forms (5,738 of them non-identity), with cross-validated recall and precision of 98.6% and 99.96% on the 759-keyword vocabulary, 99.0% and 99.98% on the 68 frequent keywords, and 97.2% and 100% for "federated learning".

Entry status. A paper is an incumbent paper if at least one author identifier has an earlier ICLR submission, with any outcome, in the paper's topic, strictly before the paper's year; otherwise it is a newcomer paper. Exact identifiers are used, never names. ICLR author identifiers were 98–99.6% e-mail-based in 2017–2020, 85% profile-based in 2021 and fully profile-based in 2025–2026, and the two kinds are not linked. Earlier histories are therefore hidden after 2020: incumbency on the all-years map falls from 18% (2020) to 0.5% (2021) and recovers to 18% (2022) and 33% (2026). This turns some true incumbents into newcomers and pulls entry contrasts towards zero. The counted R2 variant drops only 2021–2022 and leaves the entry results unchanged: +3.72 [+2.61, +4.83] and -1.74 [-2.53, -0.95] pp (n 37,312). Earlier work that sits in the held-out split is invisible to discovery histories, which also pulls towards zero. Held-out features were built on pooled histories (held-out incumbency 30.9%). Discovery incumbency is 26.5% under discovery-only histories; under pooled histories it would be 30.0%, with 1,515 discovery papers switching from newcomer to incumbent. Experience at other venues is not captured.

Probable earlier versions. 170 discovery papers are flagged as probable resubmissions: 167 have an earlier paper at centred SPECTER2 cosine $\geq \tau = 0.90$, and 3 an earlier paper with an identical normalised title. The threshold was calibrated by blind LLM judges (two Haiku runs and one Sonnet run), whose yes-rates were 10%, 48%, 90% and 100% across four cosine bins. The flags define the R1 and S4 variants and the exclusions in both distance measures.

Distances. Both distances use SPECTER2 vectors centred with a frozen discovery mean and L2-normalised. Closeness to the team's own record ($F_{\max}$) is, for each experienced author, the cosine between the paper's vector and the normalised mean of that author's earlier ICLR submissions since 2021, excluding the paper's probable earlier version and any paper at cosine ≥ 0.90; $F_{\max}$ is the maximum over experienced authors. The measure follows Bao and Li (2025). Distance from earlier ICLR work is $1-$ the mean cosine to the 10 nearest submissions from strictly earlier years, as a percentile within topic $\times$ year (Appendix E.2). The space is the same; the reference points differ.

A dropped measure. A registered measure of buzzword use (a fast-growing keyword used atypically) failed its blind validation: LLM judges rated 42% (95% CI 33–52%) of flagged uses peripheral, against 15% of matched non-flagged uses, and the registered gate required at least 50%. Its two hypotheses (H2d, H3c) were dropped before any estimate; exploratory rows that use it are reported and not interpreted.

Held-out topic maps. For each of nine alternative maps (k = 50, 75, 150 and 200, the seed refit, the year-balanced map and the maps fitted on 2017–2020, 2021–2023 and 2024–2026), the exact centres were re-derived from the frozen fits; every re-derivation reproduces all saved discovery labels. On every map the held-out topic distribution differs from discovery by no more than sampling noise, and the share of stable papers is 0.792 in both splits.

What this means for the test. The labels are consistent rather than validated against human judgement, and the identifier switch hides earlier histories. The switch pulls the entry contrasts towards zero, as does label error if it is non-differential, so neither can produce the reversal of the registered herding prediction.

C Hindsight Details

The herding reading starts from vivid waves. Section 4 shows that the number of emerging topics a map finds depends on the years used to fit it. This appendix gives the six maps in full and three further results that depend on the fitting years.

Six maps on the same papers. All maps use $k = 100$, the same 43,480 papers and the same pipeline; only the fitting years differ (Table 8). A topic is emerging if its 2024–2026 share is at least three times its 2017–2020 share, and declining if it is at most a third. Because 69% of papers are from 2024–2026, the all-years map is in effect a

Map (fitted on)	ARI	Emerging	Declining	Fit 2017/26
All years	1	29 (40.9)	13 (41.7)	0.62/0.65
Seed refit	0.69	28 (40.9)	15 (43.2)	0.62/0.65
Year-balanced	0.48	16 (42.2)	26 (38.6)	0.69/0.61
2024–2026	0.55	28 (31.0)	6 (30.2)	0.56/0.66
2021–2023	0.40	10 (20.9)	17 (31.6)	0.63/0.58
2017–2020	0.34	8 (22.3)	17 (21.9)	0.69/0.53

Table 8: Six $k = 100$ maps that differ only in their fitting years. ARI with the all-years map. Emerging: count (share of 2024–2026 papers in them, %). Declining: count (share of 2017–2020 papers in them, %). Fit: mean similarity of a year's papers to their assigned centroid. Outcome-free (E).

Year	All	Seed	Bal.	24–26	21–23	17–20
2017	0.443	0.470	0.636	0.448	0.556	0.738
2018	0.525	0.529	0.743	0.496	0.655	0.850
2019	0.582	0.589	0.808	0.520	0.745	0.888
2020	0.660	0.665	0.850	0.574	0.833	**0.908**
2021	0.703	0.713	**0.867**	0.619	0.915	0.867
2022	0.725	0.731	0.831	0.629	**0.921**	0.804
2023	0.797	0.795	0.847	0.697	0.911	0.785
2024	0.897	0.893	0.820	0.842	0.839	0.792
2025	**0.910**	**0.896**	0.731	**0.937**	0.739	0.731
2026	0.840	0.831	0.634	0.930	0.640	0.681

Table 9: Effective number of topics divided by k, by year and map (maximum in bold). Each map is most diverse in its own fitting years. Outcome-free (E); not a finding. Record: `research/topics/output/hindsight_check.json`.

present-day map: it fits 2026 papers better than 2017 papers (mean similarity to the assigned centroid 0.65 against 0.62), while the 2017–2020 map does the reverse (0.69 against 0.53). Growth compounds hindsight: the all-years map finds as many emerging topics as the map fitted on 2024–2026 alone (29 against 28). The share of 2024–2026 papers in emerging topics brackets the effect, from 22.3% on the 2017–2020 map to 40.9% on the all-years map (20.9–42.2% across the six maps), rather than pinning it down: the early map cannot represent topics born after 2020, and the maps differ in fitting corpus as well as in window (E).

Diversity is not a finding. The effective number of topics per year (as a share of k) peaks inside each map's own fitting years (Table 9): in 2020 on the 2017–2020 map, in 2022 on the 2021–2023 map, in 2025 on the 2024–2026 and all-years maps, and in 2021 on the year-balanced map. The all-years map's rise in diversity to 2025 is therefore a property of where the map was fitted, not a finding about the field.

Predictor: metric	All years	2017–2020
Size: R^2	0.281	−0.033
Size: Spearman	0.595	−0.043
Size: precision, top fifth	0.563	0.250
Size: R^2 without LLM/VLM areas	0.30	−0.09
Momentum: R^2	0.056	0.032
Momentum: Spearman	0.377	0.198
Momentum: precision, top fifth	0.420	0.395
Topic-years	446	522

Table 10: Out-of-time forecasts of two-year topic growth, test origins 2021–2024. Intervals for size: R^2 [0.217, 0.335] against [−0.082, −0.002]; Spearman [0.510, 0.672] against [−0.151, 0.063]. Chance precision is 0.20. Post hoc, outcome-free (E). Record: `research/fashion/output/ph5_forecast/summary.json` (`headline`).

Topic growth looks predictable only in hindsight. An out-of-time forecast of two-year topic growth (test origins 2021–2024; post hoc, outcome-free) shows the same asymmetry (Table 10). On the all-years map, small topics grow: topic size "predicts" growth with R^2 0.28. On the 2017–2020 map it does not (R^2 −0.03). The in-sample chance of top-fifth growth by size tercile is 40%, 21% and 2% (small, middle, large) on the all-years map, against 22%, 20% and 20% on the early map. Caveats: the early map cannot represent topics born after 2020, the maps differ in fitting corpus and eligibility timing, so the gap cannot be attributed to hindsight alone, and there is no paired test. Genuine forecasting is weak: on the early map, momentum's top-fifth picks are right for only 16 of 56 (29%) for origins 2021–2023.

The lifecycle answer depends on the map. The registered tests of whether topics "age" from method to measurement work (H1a–c) show nothing on the all-years map: method −0.14 [−1.78, +1.49] and measurement +0.50 [−0.76, +1.75] pp per log unit of prior papers (C). On the 2017–2020 map applied to 2021–2026 papers, the method and measurement estimates point the predicted way: −3.48 [−6.21, −0.76] and +3.96 [+1.35, +6.57] (counted variants, read as a lifecycle only in exploratory and unadjusted terms). The empirical share (H1c) has the wrong sign on every map. Table 11 lists every counted variant. By the registered rule all three are not supported, because the main map shows nothing; the lesson is that the answer depends on the map.

Variant	H1a (−)	H1b (+)	H1c (+)	n
All-years (main)	−0.14	+0.50	−0.31	43,096
$k = 75$	−0.10	+0.93	−0.74	43,096
$k = 150$	−0.35	+0.69	−0.13	43,096
Year-balanced^h	−0.08	+0.66	−0.75	43,096
Early map, all papers	−2.30	+2.16	−1.06	43,096
Early map, 2021–2026^h	−3.48	+3.96	−1.03	39,008
Stable papers	+0.53	+0.28	−0.37	34,144
Coherent topics	−0.20	+0.50	−0.26	42,161
Certain labels	−0.66	+0.92	−0.56	26,369
Without 2026	+0.48	−0.53	−0.16	27,685

Table 11: Topic crowding and contribution mix: every counted variant of the registered H1a (method share), H1b (measurement share) and H1c (empirical share), pp per log unit of prior topic papers; registered sign in parentheses; ^{h} hindsight variant; early map: fitted on 2017–2020. Predicted sign: 7/9, 8/9 and 0/9 variants. Outcome-free labels.

What this means for the test. A herd reading starts from the waves, and the emergence a field history shows is partly a property of when its map was fitted. For this reason every registered time-trend claim had to hold on the year-balanced map and on the 2017–2020 map applied to 2021–2026 papers. H1d, the package result, passes both; the lifecycle claims do not.

D Every Registered and Declared Test

This appendix lists every registered and declared test, including those that failed, so that the denominators are visible (Table 13). It keeps the detailed tiers and internal codes that the main text replaces with three labels. The tiers are: **A** registered and replicated; **B** found post hoc, then supported by a held-out test declared before any held-out estimate for that question was computed; **C** registered, inconclusive or not replicated; **D** discovery only; **E** descriptive. Table 12 maps every code to its name and label in the main text, and Table 14 gives the full version of the evidence summarised in Tables 2 and 3, with codes, predicted signs and notes. Figure 4 in the main text shows the gate tests by reference point.

Protocol. Hypotheses, models, the support rule and the robustness variants were registered on 2026-10-13, before any full-corpus label existed (only the 2,300 validation papers had labels, and no hypothesis had been tested on them). Earlier exploratory work on keyword popularity and title style in the same corpus came before registration and is disclosed. A hypothesis is *supported* only if its main

Code (tier)	Main-text name (section)	Main-text label
Registered 2026-10-13 (H4-FL: 2026-10-17)		
H1a–c (C)	Topic crowding and the contribution mix (App. C)	not supported
H1d (A)	Package shift: prior FM adoption and new-architecture papers (§4)	REPLICATED
H1e (C)	Prior FM adoption and empirical or evaluation papers (§4)	not supported (inconclusive)
H2a (C)	Flocking: incumbent share in hot topics (§5)	inconclusive
H2b, H2b′ (A)	Newcomer papers and method papers (§5)	REPLICATED
H2c, H2c′ (A)	Newcomer papers and measurement papers (§5)	REPLICATED
H3a (C)	Crowding and acceptance (§6)	uninformative
H3b (C)	Crowding × method paper (appendix only)	inconclusive
H4-FL (C)	Federated-learning gap (appendix only)	not confirmed
Post hoc; each held-out test declared before any held-out estimate for that question was computed		
PH5 E1 (B)	Far jump (§6)	HELD-OUT SUPPORTED
PH5 E4 (B)	Same author, +0.1 cosine closer (§6)	HELD-OUT SUPPORTED
Teams H1 (B)	One topic insider vs none (§6)	HELD-OUT SUPPORTED
Premia *b*, D0 (B)	Topic edges are real but regress (§6)	HELD-OUT SUPPORTED
PH5 P1b (C)	Least-distant fifth vs middle: the field's record (§6)	uninformative
PH5 P7 (C)	Same author, +10 percentile points of distance (§6)	not supported
A1 (B weak)	Emerging vs Mainstream keywords (§6)	EXPLORATORY (demoted)
PH6 (B weak)	Acceptance after a keyword's peak (§6)	EXPLORATORY (demoted)
PH7 (D)	Branded titles (§6, App. G)	EXPLORATORY
Exploratory, no held-out test		
Planned E1; fmwave (D)	FM premium and its change with prior adoption (§6)	EXPLORATORY
Planned E2; PH3 (D)	Newcomer acceptance gap; same author in a new topic (§6)	EXPLORATORY
Teams per head (D)	Insider vs experienced outsider, per head (§6)	EXPLORATORY
Profiles (E)	Entering a topic, not ICLR (§5, App. E)	EXPLORATORY
Distance (E)	Newcomer papers farther from earlier ICLR work (§5)	EXPLORATORY
Twins, FM bins, kwtiming profiles (E)	Same-year twins; first-time teams later onto FMs and Emerging keywords (§5)	EXPLORATORY
Hindsight maps (E)	Emerging topics on six maps (§4)	EXPLORATORY

Table 12: From internal code and tier to the name and label used in the main text. A1 and PH6 were supported on held-out papers only weakly and are demoted to exploratory in the main text: A1 does not survive Holm across the six secondary held-out tests (0.063), and the PH6 wave test leans on the six keywords inspected first.

estimate has the predicted sign, its Holm-adjusted p is below 0.05 within its family (Holm, 1979), at least 75% of its counted variants have the predicted sign, and both hindsight variants (year-balanced map; 2017–2020 map on 2021–2026 papers) have the predicted sign. Otherwise the verdict is "significant in the opposite direction" (Holm $p < 0.05$ with the wrong sign), "equivalent to null" (90% CI inside the smallest effect of interest) or "inconclusive". Discovery Holm families were {H1a–e}, {H2a–c} and {H3a–b}. Linear probability models use topic and year (or topic × year) fixed ef-

fects with standard errors clustered by topic (100 clusters). Outcomes were readable only through a sealed loader, which ran once inside the frozen run (18 seconds). Nine adversarial review rounds preceded the freeze; two blinding incidents are disclosed (a reviewer saw one exploratory estimate on the failed buzzword measure, and the analyst once read the value sets of outcome columns).

Held-out tests. Held-out tests were registered on 2026-10-18 and run once (16:39–16:42 UTC). Every test is one-sided at $\alpha = 0.05$: $p_1 = p/2$ if the estimate has the predicted sign, otherwise $1 - p/2$, where p is the two-sided p of the discovery method; the exception is premia b, whose p_1 is the intersection-union p for $0 < b < 1$ (0.003). Holm families were {H1d}, {H2b′, H2c′}, {H3a}, {H4-FL} and the post hoc primary pair {E1, P1b}. Six post hoc secondary tests (E4, P7, teams H1, A1, premia b, premia D0) are each one-sided at 0.05 and unadjusted, so one false positive among them would not be surprising; their Holm values hold within their own module only (teams H1 about 0.17 across modules, A1 about 0.18). Under Holm across the six secondary tests, the adjusted one-sided p, computed from the exact one-sided p (0.003084, 0.005125, 0.006184, 0.017427, 0.031489, 0.21699), are 0.019 (premia b), 0.026 (D0), 0.026 (E4), 0.052 (teams H1), 0.063 (A1) and 0.22 (P7): E4 and both premia tests survive this family-wise correction, teams H1 and A1 do not. The main text keeps the insider association as held-out supported with that caveat, and demotes A1 to exploratory. The PH6 wave test was declared on 2026-10-28 and run once; it is exploratory in the main text, because it leans on the six keywords inspected first. The PH7 branded-title test was declared on 2026-11-04 and run once (Appendix G). H2b′ and H2c′ reverse the registered herding predictions, which failed in discovery; they were registered after discovery and before any held-out data existed.

Discovery details. H1a–c, H1e: Holm 1.0; predicted sign in 7/9, 8/9, 0/9 and 4/9 counted variants (H1c also fails both hindsight variants). H1d: Holm 0.0014; 9/9 variants (-6.3 to -14.2). H2a: Holm 0.46; 8/10 variants with the predicted sign, but the year-balanced hindsight variant has the wrong sign (+1.05), as does $k = 150$ (+0.37); the two early-map variants are -1.62 (p 0.03) and -1.55 (p 0.045); TOST p 0.10. H2b and H2c: wild cluster restricted bootstrap $p < 1/9{,}999$, Holm < 0.001 with the wrong sign; all 10 counted variants share the observed sign (+2.16 to +3.87; -1.81 to -0.86), as do both hindsight variants (+3.10 and +2.16; -1.41 and -0.86). With log team size, H2c shrinks to -0.54 [-1.11, +0.04] (p 0.07) while H2b stays at +3.92; by era, H2c is concentrated in 2024–2026 (-2.0, against -0.5 n.s. in each earlier era). H3a: Holm 0.20; all 11 counted variants negative (-2.53 to -1.31; logit average marginal effect -1.64); supplementary variants are negative and some nominally significant (team size -2.04, fine topics -2.42, withdrawals after review counted as rejections -2.12, newcomer control -2.99), but they are unadjusted and outside the verdict; TOST p 0.75. H3b: Holm 0.53; 10/11 variants negative; TOST p 0.14. No non-supported hypothesis is equivalent to null.

Held-out details. H1d: 9/9 variants negative (-5.4 to -15.2); the weakest, the prospective early-map variant, is -5.4 (two-sided p 0.24). H2b′ and H2c′: 10/10 variants, both hindsight variants pass; two-way clustering agrees (Holm p_1 0.0010 and 0.0009); the stress sensitivity gives Holm p_1 0.0075 for both; no variant is significant against the prediction. With log team size, H2c′ shrinks to -0.98 (two-sided p 0.10) while H2b′ stays at +3.64, as in discovery. H3a: 9/11 variants negative; the hindsight requirement fails (early map on 2021–2026 papers +1.44); TOST p 0.61. H4-FL: yearly held-out gaps of -20.0, -16.9, +6.4, -3.6, -11.3 and -0.2 pp (2021–2026, on 13–36 papers a year); the discovery pattern of about -11 to -15 pp in every year since 2023 does not appear.

Reading rules. Four supported tests were underpowered at the discovery effect: H1d (MDE80 12.0 against -7.98), H2c′ (2.4 against -1.70), teams H1 ($\approx$4.5 against +3.42) and A1 ($\approx$4.3 against +2.67). They confirm direction, not size; among underpowered tests, the significant ones tend to overstate the effect. A non-supported registered test whose estimate is below its MDE80 (H3a, H4-FL, P1b) is uninformative, not evidence of no association. A1's 95% interval includes 0 because a one-sided 5% test corresponds to a 90% interval.

Further notes to Table 14. Row 6 (distance from earlier ICLR work): outcome-free, unregistered and added during writing; among method papers the gap is +5.24 [+4.16, +6.31]; "own work excluded" means that every earlier paper sharing an author with the paper is removed from its reference

set (Appendix E.2). Row 7: the FM gaps compare first-time teams with returning topic newcomers; no gap is detectable at 0–5% (-0.95) or $\geq$50% (-1.96) prior adoption, or between returning topic newcomers and incumbents in any bin (-2.58 to +2.87); the keyword contrast compares first-time with returning teams and has no team-size control. Row 11: with keywords as units the held-out slope is +0.16 [-0.34, +0.66] (unregistered); empirical-Bayes-shrunk edges predict better than zero (D0 1.71×10^{-3}, B). Row 17: estimated at fixed team size and number of experienced authors; underpowered at the discovery effect; its Holm value holds within its module only ($\approx$0.17 across modules; 0.052 under Holm across the six secondary held-out tests); an early-map insider gives +3.0 and a keyword-defined insider +0.0.

PH6 variants. The wave test is post hoc: the question came from viewing the discovery per-year gaps of six keywords, and its wave set is retrospective by construction. Table 15 shows that it leans on those six keywords: without them the held-out estimate is -1.7 (p_1 0.33). Vision transformer has the largest drop in both splits, and federated learning itself does not decline in the held-out papers (+3.5 pp after against up to its peak; discovery -5.9), consistent with the failed H4-FL trend. For these reasons the main text reports the wave test as exploratory, in one sentence.

Unregistered held-out rows. The held-out run also produced 17 rows that were not registered (Table 16). They are two-sided and unadjusted, and they are reported, not interpreted. One quoting rule follows from them, in either direction: a discovery reading that rests on an unregistered contrast is quoted only with its held-out and pooled rows (P2 and E2 in Appendix F).

Pooled re-run (confirms nothing). After the held-out run, every analysis was re-run on the pooled corpus (54,579 papers). It confirms nothing: about 80% of each pooled sample is discovery papers, so agreement with discovery is expected. Pooling also changes discovery features: hot status changes for 8.1% of discovery papers, discovery incumbency rises from 26.5% to 30.0%, probable earlier versions rise from 170 to 224, and crowding rises by about 0.22 log units. Table 17 gives the main rows. Every registered estimate keeps its discovery sign and descriptive verdict except H3b, which moves to zero (TOST p 0.034, labelled "equivalent to null" only descriptively); H1a -0.59, H1b +0.81, H1c -0.24, H1e -0.6 and H2a -0.85 remain not supported or inconclusive. Four topics pass Benjamini–Hochberg correction in the pooled area analysis (clinical ML -16.2, federated learning -12.8, graph representation learning -9.7, robot learning +12.1); no keyword does. The pooled P1b rests mainly on the discovery papers.

What this means for the test. Read by reference point, the denominators line up with the main text. The package convergence (H1d) replicated but does not discriminate between the routes. The registered imitation-side tests either failed in the opposite direction (H2b, H2c) or were inconclusive (H2a; H3a, which predicted a crowding penalty and whose intervals exclude premiums above +0.32 and +2.19 pp per log unit: no detectable premium). Among the post hoc gate patterns taken to held-out papers, those referenced to the team's own record held (E1, E4, teams H1) and topic edges regressed (premia b, D0); the timing tests held only weakly (A1, PH6) and are exploratory in the main text; the branded-title premium (PH7) was not confirmed; and those referenced to the field or to one vivid case did not hold (P1b, P7, H4-FL).

Test (predicted sign; held-out n)	Discovery [95% CI]	Held-out [95% CI]	p_1	MDE80	Verdict (tier)
Field: registered 2026-10-13; H1d also registered for held-out papers					
H1a method share ∼ crowding (−)	−0.14 [−1.78, +1.49]	not tested			not supported (C)
H1b measurement share ∼ crowding (+)	+0.50 [−0.76, +1.75]	not tested			not supported (C)
H1c empirical share ∼ crowding (+)	−0.31 [−1.29, +0.67]	not tested			not supported (C)
H1d new-architecture share ∼ prior FM adoption, 0→100% (−; 10,978)	−7.98 [−12.14, −3.82]	−11.0 [−18.1, −3.9]	0.0013	12.0	supported (A)
H1e empirical or evaluation share ∼ prior FM adoption (+)	−0.63 [−3.64, +2.38]	not tested			not supported (C)
Entry: registered 2026-10-13; reversed H2b′ and H2c′ registered for held-out papers after discovery					
H2a incumbent share ∼ hot topic (−)	−0.73 [−2.68, +1.22]	not registered[u]		3.6[f]	inconclusive (C)
H2b method paper ∼ newcomer paper (−); reversed H2b′ (+; 10,718)	+3.71 [+2.61, +4.80]	+3.60 [+1.48, +5.71]	0.0007	2.8 (1.7[f])	failed; reversal supported (A)
H2c measurement paper ∼ newcomer paper (+); reversed H2c′ (−; 10,718)	−1.70 [−2.46, −0.93]	−2.10 [−3.35, −0.85]	0.0005	2.4 (1.1[f])	failed; reversal supported (A)
Gate: registered 2026-10-13 (H3a also for held-out papers); H4-FL registered 2026-10-17, before any held-out data					
H3a acceptance ∼ crowding, per log unit (−; 8,185)	−1.68 [−3.68, +0.32]	−1.51 [−5.20, +2.19]	0.21	4.8	inconclusive; uninformative (C)
H3b acceptance ∼ crowding × method paper (−)	−0.36 [−1.51, +0.79]	not registered[u]		2.4[f]	inconclusive (C)
H4-FL federated-learning keyword, level (−; 134 FL papers)	−11.3 [−17.9, −4.8]	−4.9 [−15.4, +5.7]	0.14	14.5	not supported; uninformative (C; discovery D)
H4-FL trend, pp per year (−)	−3.0 [−6.4, +0.5]	+3.2 [−3.3, +9.6]	0.88	n.c.	not supported, opposite sign (C)
Gate: post hoc in discovery (declared before run); held-out tests declared 2026-10-18 (PH6: 2026-10-28; PH7: 2026-11-04)					
E1 far jump: $F_{\max}$ p10 − p50 (−; 5,344)	−5.44 [−7.10, −3.78]	−6.09 [−9.32, −2.86]	0.0025	4.9 (0.89)	supported, primary (B)
P1b field distance: least-distant fifth − middle fifth (−; 8,010)	−3.27 [−4.93, −1.60]	−0.13 [−3.50, +3.25]	0.47	5.4 (0.47)	not supported; uninformative (C; discovery D)
E4 within-author closeness, per +0.1 cosine (+; 4,311)	+0.55 [+0.21, +0.90]	+1.06 [+0.23, +1.89]	0.006	n.c.	supported, secondary (B)
P7 within-author field distance, per 10 percentile points (+; 4,810)	+0.39 [+0.09, +0.69]	+0.34 [−0.47, +1.16]	0.22	≈1.0–1.1[d]	not supported (C)
Teams H1: one topic insider vs none (+; 6,243)	+3.42 [+1.18, +5.66]	+3.88 [+0.28, +7.47]	0.017	≈4.5[d]	supported, secondary (B)
A1 Emerging − Mainstream keyword (+; 5,741)	+2.67 [+0.76, +4.58]	+3.25 [−0.18, +6.69]	0.031[n]	≈4.3[d]	supported, weak: CI includes 0 (B weak)
Premia b: 2024–2026 topic edge on 2018–2022 edge ($0 < b < 1$; 75 units)	0.38 [0.18, 0.58]	0.60 [0.32, 0.88]	0.003[i]	n.c.	supported, secondary (B)
Premia D0: EB-shrunk edges predict better than zero, $\times 10^{-3}$ (+)	none (new test)	1.71 [0.42, 3.01]	0.005	n.c.	supported, secondary (B)
PH6 waves: gap after − up to a keyword's share peak (−; 17 waves)	−5.7 [−8.5, −2.9][t]	−5.2 [−11.1, +0.7]	0.043	7.4	supported, only just: CI includes 0 (B weak)
PH7 branded title (D2) vs other, class, FM and team-size controls (+; 8,185; D1, D3, D4 in Table 23)	+1.74 [+0.37, +3.11][t]	+1.76 [−0.82, +4.34]	0.089	3.2[a]	not supported; uninformative (D)

Table 13: Every registered and declared test. Estimates in pp within topic and year (H1d: change in the new-architecture share for prior-year FM adoption from 0 to 100%; keyword and wave rows: against all other papers of the same year). Discovery intervals ±1.96 SE; the held-out H1d interval uses $t(G-1)$ quantiles. p_1: held-out one-sided p; Holm-adjusted, it is unchanged except for H2b′ and H2c′ (0.0010 each) and E1 (0.0050). MDE80: held-out minimum detectable effect at 80% power, registered unless marked, with power at the discovery effect for E1 and P1b in parentheses; [f] discovery MDE80 from the frozen dry run; [d] derived post hoc (2.49×SE), with no reading rule; n.c. not computed. [u] unregistered held-out row (Table 16). [n] null-frame p_1 0.020; two-sided p 0.063. [i] intersection-union p for $0 < b < 1$. [t] not a test (PH7: title style had been explored before registration). [a] computed analytically in the declared run (2.49×SE). Tiers as at the start of this appendix; main-text labels in Table 12; reading rules in the text.

#	Observable [code]	Predicted: imitation / continuity	Discovery [95% CI]	Held-out [95% CI]; MDE80 or power	Tier	Reading
(i) The field: are the waves fashions?						
1	Emerging topics: all-years vs 2017–2020 map [hindsight vs P]	· / ·	29 (40.9%) vs 8 (22.3%)	–	E	emergence depends on the map
2	Prior-year FM adoption 0→100% → new-architecture paper [P, H✓; H1d]	− / − (lock-in)	−7.98 [−12.14, −3.82]	−11.0 [−18.1, −3.9]; MDE80 12.0[a]	A	convergence; does not discriminate
3	Twins per paper per 1,000 submissions, 2019→2026 [same-year]	+ / ·	0.32→0.19	–	E	volume, not piling on
(ii) Entry: what topic newcomers write						
4	Newcomer → method paper (base 75.0%) [P, H✓; H2b, H2b′]	− (registered) / +	+3.71 [+2.61, +4.80]	+3.60 [+1.48, +5.71]; MDE80 2.8	A (rev.)	against imitation[h]
5	Newcomer → measurement paper (base 7.7%) [P, H✓; H2c, H2c′]	+ (registered) / −	−1.70 [−2.46, −0.93]	−2.10 [−3.35, −0.85]; MDE80 2.4	A (rev.)	against imitation; partly team size[g]
6	Newcomer → distance from earlier ICLR work [P]	− / + (incumbents' own work)	+5.90 [+4.95, +6.85]; own work excluded +4.33 [+3.44, +5.22]	–	E	against imitation[i]
7	First-time vs returning team → Emerging keyword; first-time team vs returning topic newcomer → FM paper at 5–15 / 15–30 / 30–50% adoption, 2023–2026 [P]	+ / − (absorptive capacity)	−3.74 [−5.14, −2.35]; −4.58 / −8.96 / −14.71	–	E	later onto trend and package
8	Hot topic → incumbent share (flocking) [P, H×; H2a]	− / ·	−0.73 [−2.68, +1.22]	not registered	C	inconclusive
(iii) The gate: following and timing						
9	Crowding, per log unit → acceptance [P, H×; H3a]	+ if favoured (registered: −) / ·	−1.68 [−3.68, +0.32]	−1.51 [−5.20, +2.19]; MDE80 4.8	C	no detectable premium[b]
10	Branded title (D2) → acceptance [explored before registration; PH7]	+ if favoured / ·	+1.74 [+0.37, +3.11] (not a test)	+1.76 [−0.82, +4.34]; MDE80 3.2	D	small premium, not confirmed[j]
11	Topic edge 2018–2022 → edge 2024–2026, slope b [P; premia b]	$b > 0$ if outcome imitation is favoured / ·	0.38 [0.18, 0.58]: discovery 2024–2026 edges on discovery 2018–2022 edges	0.60 [0.32, 0.88]: held-out 2024–2026 edges on discovery 2018–2022 edges	B	real but regressing
12	Emerging minus Mainstream keyword [P; A1]; after vs up to a keyword's peak, 17 waves [R; PH6]	+ / + (priority); 0 or − / −	+2.67 [+0.76, +4.58]; −5.7 [−8.5, −2.9]	+3.25 [−0.18, +6.69]; −5.2 [−11.1, +0.7] (MDE80 7.4)	B, weak; main text D	timing open; exploratory[c]
13	Federated-learning keyword gap [R; H4-FL]	· / ·	−11.3 [−17.9, −4.8]	−4.9 [−15.4, +5.7]; MDE80 14.5	D/C	not confirmed[d]
(iv) The gate: distance from the field's and from the team's own record						
14	Field distance: least-distant vs middle fifth [P, T✓; P1b]	+ if favoured (declared: −) / + (field record)	−3.27 [−4.93, −1.60]	−0.13 [−3.50, +3.25]; power 0.47	C	undetermined; does not discriminate[e]
15	Far jump from own record: $F_{\max}$ p10 − p50 [P, T✓; PH5 E1]	· / −	−5.44 [−7.10, −3.78]	−6.09 [−9.32, −2.86]; MDE80 4.9	B (prim.)	for continuity[f]
16	Same author, per +0.1 cosine closer [P, T✓; PH5 E4]	· / +	+0.55 [+0.21, +0.90]	+1.06 [+0.23, +1.89]	B (sec.)	for continuity
17	One topic insider vs none [P; teams H1]	· / +	+3.42 [+1.18, +5.66]	+3.88 [+0.28, +7.47]; MDE80 ≈4.5	B (sec.)	status-compatible

[a] Underpowered at the discovery size: direction, not size. [b] Registered H3a predicted a penalty (−); one-sided held-out p 0.21; the hindsight requirement fails held-out (early map on 2021–2026 papers +1.44). Penalties below about 5 pp per log unit cannot be told from zero; premiums above +0.32 / +2.19 pp per log unit lie outside the intervals. [c] 'Early' means early in a keyword's life, not in the submission window (cf. Jung et al., 2025). Imitation predicts + for Emerging if momentum is favoured, and after the peak 0 (popularity) or − (momentum). The Emerging contrast's two-sided interval includes 0 (MDE80 ≈4.3); old rising keywords +3.9 (nominal). Discovery wave estimate descriptive, not a test; wave set retrospective; held-out −1.7 (p 0.33) without the six keywords inspected first. In discovery the gap falls before the peak, and rising against flat years of the same keyword differ by +0.9 [−0.8, +2.7]. Exploratory in the main text: A1 gives Holm 0.063 across the six secondary held-out tests, and the wave test leans on the six keywords inspected first. [d] Held-out trend of opposite sign (+3.2 pp per year). [e] Uninformative. Rows 14–17 differ in population and unit: compare replication status, not size. [f] Newcomer control −5.0; link-removed −5.39; early map −5.2; correlation with newcomer flag −0.57. [g] With log team size: −0.54 [−1.11, +0.04] in discovery, −0.98 held-out; row 4 unchanged (+3.92 in discovery, +3.64 held-out); supplementary to A. [h] Rows 4–5: all 10 counted variants agree, and the t-statistics (6.7, −4.4) dwarf the size-failure inflation factors (1.31, 1.22). [i] Raw scale: +1.35 [+1.16, +1.55] hundredths of cosine distance, 0.22 of a within-cell standard deviation. [j] Title style had been explored before registration, so the discovery estimate is not a test; declared held-out test one-sided p 0.089, not supported and uninformative; raw within-year gaps +3.31 and +3.97 (Appendix G).

Table 14: The herding account tested at three points, in full: each observable, its internal code and the sign predicted by imitation and by continuity (in parentheses: its mechanism or reference point, or a named alternative: lock-in, priority; ·: none). Percentage points unless the row states otherwise (row 1: counts; row 3: twins per paper per 1,000 submissions; row 6: percentile points; row 11: slope b), 95% intervals, within topic and year (keyword rows and row 11: against the same year's other papers). Held-out tests are one-sided and confirm direction, not size, so a supported test can have an interval that includes 0. Codes: P only earlier data; same-year: twins are concurrent papers, so not only earlier data; H✓/H× registered hindsight variants pass/fail; R retrospective by construction; T✓ held after link removal. Tiers as at the start of this appendix; rev.: reversed registered prediction; prim./sec.: declared primary/secondary held-out test (six secondary tests in all, among them rows 11, 12 (Emerging), 16 and 17, one-sided and unadjusted; under Holm across the six they give 0.019, 0.063, 0.026 and 0.052); D/C: discovery-only gap whose registered held-out test was not supported. The continuity predictions were written after the tests had run. Main-text labels in Table 12.

Split (waves)	D	95% interval	p_1	MDE80
Held-out, declared test (17)	−5.2	[−11.1, +0.7]	0.043	7.4
Held-out, without six inspected (11)	−1.7	[−9.7, +6.3]	0.33	10.1
Held-out, without FL (16)	−6.2	[−12.5, +0.1]	0.033	7.9
Discovery, not a test (17)	−5.7	[−8.5, −2.9]	<0.001	3.6
Discovery, without six inspected (11)	−3.7	[−7.5, +0.2]	0.034	4.8

Table 15: PH6: weighted mean over waves of a keyword's acceptance gap after minus up to its share peak, pp; 2,000 within-year permutations. Held-out papers before and after peaks: 454 and 1,022.

Row (sign of interest)	Discovery [95% CI]	Held-out [95% CI]	p
H2a incumbent share $\sim$ hot topic $(-)$	-0.73 [-2.68, +1.22]	-2.55 [-5.59, +0.49]	0.099
H3b acceptance $\sim$ crowding $\times$ method $(-)$	-0.36 [-1.51, +0.79]	+1.34 [-1.26, +3.93]	0.31
H4-FL recent level, 2024–2026	-13.5 (p 0.007)	-3.6 [-17.2, +9.9]	0.52
H4-FL level adjusted for class, FM and team size	-10.5	-3.4	0.42
E2 near: $F_{\max}$ p90 $-$ p50	+1.75 [-0.17, +3.66]	+2.34 [-1.01, +5.70]	0.17
E3 far jump, hot minus other topics	-0.35 [-3.13, +2.42]	-3.09 [-9.63, +3.45]	0.35
P1a field distance, most-distant fifth $-$ middle fifth	+0.44 [-1.41, +2.28]	+2.54 [-1.40, +6.48]	0.20
P2 any independent same-year twin	+0.68 [-0.84, +2.20]	+3.69 [+0.90, +6.48]	0.010
P3 twin association, high minus low field distance	+0.18 [-3.46, +3.83]	+4.84 [-2.27, +11.95]	0.19
P4 twin association, hot minus other topics	-1.49 [-4.51, +1.52]	+1.47 [-3.65, +6.59]	0.63
P5 (Q5 $-$ Q3) field distance, newcomer minus incumbent papers	+1.32 [-2.63, +5.28]	-9.75 [-17.38, -2.12]	0.030
P6 (Q5 $-$ Q3) field distance, mature minus young topics	+4.69 [+0.21, +9.18]	+10.74 [+1.95, +19.53]	0.030
Teams H2 concavity: (1 vs 0) $-$ (2+ vs 1) insiders	+2.95 [-1.41, +7.30]	+4.47 [-2.24, +11.18]	0.19
Premia b, all-years-map topics (62 units)	0.35 [0.17, 0.54]	0.51 [0.23, 0.79]	0.0004^{i}
Premia b, keywords (35 units)	0.53 [0.10, 0.95]	+0.16 [-0.34, +0.66]	0.26^{i}
Premia b, areas (20 units)	0.20 [-0.49, 0.89]	0.71 [0.09, 1.33]	0.17^{i}
Premia D = MSE(raw) $-$ MSE(EB), $\times 10^{-3}$ (descriptive)	none	0.37 [-1.31, +2.06]	0.33^{o}

Table 16: Unregistered held-out rows: reported, not interpreted. Two-sided, unadjusted p; ^{i} intersection-union p; ^{o} one-sided. Estimates in pp unless noted. P5 and P6 change sign or size between splits and are not interpreted; they are listed so that the count of 17 rows is complete.

Analysis	Discovery	Pooled
H1d (0$\rightarrow$100% FM)	-8.0	-8.9
H2b method $\sim$ newcomer	+3.71	+3.46
H2c measurement $\sim$ newcomer	-1.70	-1.65
H3a per log unit	-1.68	-1.72
H3b	-0.36	-0.03
FM papers, acceptance	+4.62	+5.61
Newcomer papers, acceptance	-10.2	-10.2
New arch. $\times$ prior FM	+8.3	+9.4
Team size, per doubling	+8.8	+8.8
Newcomer, author FE	-3.1	-3.1
FL keyword, level	-11.3	-9.9
FL keyword, trend per year	-3.0	-1.9
E1 far jump	-5.44	-5.53
E2 near	+1.75	+2.10
P1b least-distant fifth	-3.27	-2.97
P2 any twin	+0.68	+1.86
E4 within-author closeness	+0.55	+0.78
P7 within-author field distance	+0.39	+0.43
Teams H1 one insider	+3.42	+2.90
A1 Emerging $-$ Mainstream	+2.67	+3.10
Premia b	0.38	0.45

Table 17: Pooled re-run (exploratory; confirms nothing). pp unless noted. Pooled Holm or q values: H1d 3×10^{-4}; H3a 0.098; FL level q 0.14 (trend p 0.18); E2 0.036; P2 0.063. The discovery outputs remain the registered record.

E Entry Details

Section 5 reads the registered entry result as a division of labour. This appendix gives the two descriptive analyses behind that reading: what the newcomer contrast is made of, and how far newcomer papers sit from earlier ICLR work. Both were added during writing, read no outcome, are unregistered and are descriptive (E).

E.1 What the newcomer reversal is made of

Profiles. The population is the registered entry population (discovery papers 2019–2026 with an author identifier, all-years map; 42,325 papers). Each paper has one of three profiles: a *first-time team*, in which no author has any earlier ICLR submission (11,112 papers, 26.3%; median team size 3); a *returning topic newcomer*, new to the topic but with an author who has an ICLR submission in another topic (19,765, 46.7%; median 5); and a *topic incumbent* (11,448, 27.0%; median 5). Contrasts are linear probability models within topic × year, with standard errors clustered by topic (CR1). The single newcomer contrast in this code reproduces the registered estimates (+3.71 and −1.70). Script `research/fashion/paper/analysis/newcomer_profiles.py`; outputs `research/fashion/output/paper_profiles/`.

Method. The method surplus is mainly a topic-entry pattern (Table 18). Returning topic newcomers, who already have ICLR work elsewhere, write method papers more often than incumbents (+3.43), at the same median team size and also with team size controlled (+3.62); first-time teams do not differ detectably from them (+0.92). By era (Table 19), first-time teams write method papers more often than incumbents in every era (+6.61, +5.00, +3.74), so the surplus predates the FM wave; the returning-minus-incumbent contrast is positive in every era, and its 2021–2023 interval, in the era of the identifier switch, includes 0.

Measurement. For first-time teams the measurement deficit is accounted for by team size (in the statistical sense): −2.24 against incumbents without, +0.09 with a team-size control. For returning topic newcomers part of it remains with team size (−0.81). The deficit is a 2024–2026 pattern: first-time teams against incumbents −2.75 [−4.07, −1.44] and returning topic newcomers against incumbents −1.71 [−2.57, −0.85], against intervals that include 0 in both earlier eras. In the FM era, benchmark and evaluation work sits with topic incumbents and larger teams; team size is the only resource measured, and we make no claim about compute.

Contrast (pp)	Without team size	With $\log_2$ team size
Method paper		
Returning topic newcomer − incumbent	+3.43 [+2.31, +4.55]	+3.62 [+2.44, +4.79]
First-time team − returning topic newcomer	+0.92 [−0.44, +2.29]	+1.40 [−0.15, +2.95]
First-time team − incumbent	+4.35 [+2.82, +5.89]	+5.02 [+3.12, +6.91]
Measurement paper		
Returning topic newcomer − incumbent	−1.47 [−2.23, −0.70]	−0.81 [−1.46, −0.16]
First-time team − returning topic newcomer	−0.78 [−1.51, −0.05]	+0.90 [+0.15, +1.65]
First-time team − incumbent	−2.24 [−3.24, −1.25]	+0.09 [−0.63, +0.80]
Raw shares, method / measurement (%)		
First-time team	77.56 / 5.01	
Returning topic newcomer	75.12 / 8.47	
Topic incumbent	72.48 / 9.01	

Table 18: The newcomer contrast by profile, within topic × year, 2019–2026 (42,325 papers; 95% intervals). Descriptive and unregistered (E).

Contrast (pp)	2019–2020	2021–2023	2024–2026
Method paper			
Returning − incumbent	+4.43 [+0.01, +8.84]	+2.28 [−0.29, +4.84]	+3.61 [+2.36, +4.86]
First-time − returning	+2.19 [−1.35, +5.72]	+2.73 [+0.57, +4.89]	+0.12 [−1.53, +1.78]
First-time − incumbent	+6.61 [+2.11, +11.12]	+5.00 [+2.12, +7.89]	+3.74 [+1.94, +5.53]
Measurement paper			
Returning − incumbent	−0.34 [−1.68, +1.01]	−0.30 [−1.49, +0.89]	−1.71 [−2.57, −0.85]
First-time − returning	−0.36 [−1.78, +1.06]	−0.41 [−1.43, +0.61]	−1.04 [−2.08, −0.00]
First-time − incumbent	−0.70 [−2.34, +0.94]	−0.71 [−1.93, +0.51]	−2.75 [−4.07, −1.44]
Papers	3,318	8,881	30,126

Table 19: The profile contrasts by era, without a team-size control (95% intervals). Returning: returning topic newcomer. The 2021–2023 era contains the identifier switch, which hides earlier histories. Descriptive and unregistered (E).

Package uptake. Figure 5 shows the FM-paper share gap by the topic's prior-year FM adoption among 2023–2026 submissions, with class and team-size controls. First-time teams write FM papers less often than returning topic newcomers at 5–15%, 15–30% and 30–50% prior adoption (−4.58 [−6.55, −2.60], −8.96 and −14.71 [−19.82, −9.61]); no gap is detectable at 0–5% (−0.95) or ≥50% (−1.96), or between returning topic newcomers and incumbents in any bin (−2.58

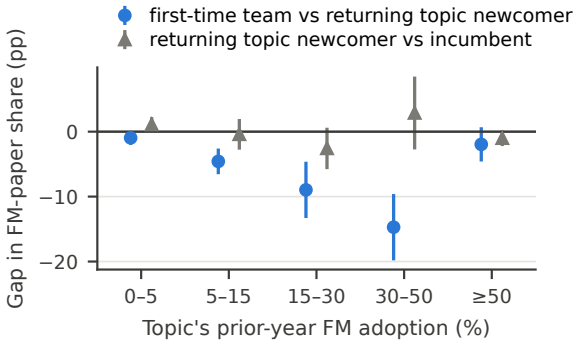


Figure 5: FM-paper share gap by the topic's prior-year FM adoption, 2023–2026 submissions, class and team-size controls: first-time teams vs returning topic newcomers (blue circles); returning topic newcomers vs incumbents (grey triangles); 95% intervals. Exploratory (E). Uses no acceptance data.

to +2.87). Absorptive capacity and network diffusion both predict this order.

E.2 Distance from earlier ICLR work

The follow-on reading of herding predicts that topic newcomers write work that sits close to what the topic has already produced. This subsection gives the full comparison behind Section 5.

Definition. Vectors are SPECTER2 embeddings of title and abstract, centred with the frozen discovery mean and L2-normalised. A paper's distance from earlier ICLR work is $1-$ the mean cosine to its 10 nearest discovery submissions from strictly earlier years (any topic), with the paper's own chain of probable earlier versions removed from the reference set. It is then ranked within the paper's topic $\times$ year cell over all submissions in the cell and reported in percentile points. The measure reimplements the one used by the gate's field-distance tests (P1b, P7) at $K = 10$, without importing that module. The population is the entry population (2019–2026 papers with an author identifier; 42,325 papers, 786 cells, 100 topics). The contrast is identified by the 575 cells that contain both newcomer and incumbent papers (39,500 papers). Estimates come from linear models with topic $\times$ year fixed effects and standard errors clustered by topic (CR1, t with $G - 1$ degrees of freedom). In the own-work-excluded variant, every earlier paper that shares an author key with the paper is removed from its reference set; author keys are used only as join keys and never written out. The script and its outputs are committed with the study's code.

Specification	Gap [95% CI]	N
Main	+5.90 [+4.95, +6.85]	42,325
Method papers	+5.24 [+4.16, +6.31]	31,764
Method papers, new architectures excluded	+5.00 [+3.86, +6.14]	27,910
Log_2 team size	+5.59 [+4.64, +6.54]	42,325
Class, FM and team-size controls	+5.76 [+4.82, +6.70]	42,325
Method papers, team size	+5.19 [+4.12, +6.26]	31,764
2019–2022	+4.95 [+2.32, +7.58]	8,327
2023–2026	+6.01 [+5.03, +6.98]	33,998
2019–2022, team size	+4.68 [+2.15, +7.20]	8,327
2023–2026, team size	+5.69 [+4.71, +6.67]	33,998
Later minus earlier era	+1.06 [−1.64, +3.75]	42,325
Own work excluded	**+4.33 [+3.44, +5.22]**	42,325
Own work excluded, team size	+4.01 [+3.13, +4.89]	42,325
Own work excluded, method papers	+3.71 [+2.70, +4.72]	31,764
Raw distance $\times 100$	+1.35 [+1.16, +1.55]	42,325
Raw $\times 100$, team size	+1.29 [+1.09, +1.49]	42,325
Raw $\times 100$, method papers	+1.19 [+0.97, +1.42]	31,764
Without 2021	+5.87 [+4.92, +6.83]	39,965
Incumbency ignoring own earlier versions	+5.84 [+4.88, +6.79]	42,325
Papers without an earlier version	+5.81 [+4.84, +6.77]	42,155
Plain definition	+6.63 [+5.78, +7.48]	42,325

Table 20: Newcomer minus incumbent papers, distance from earlier ICLR work in percentile points within topic $\times$ year (raw rows: hundredths of cosine distance). Plain definition: uncentred vectors, no earlier-version exclusion (correlation with the main measure 0.74; 0.77 for percentiles). Outcome-free, unregistered (E).

Main result. Within topic and year, newcomer papers sit +5.90 percentile points [+4.95, +6.85] farther from earlier ICLR work than incumbent papers (mean percentile 51.4 against 46.1; E). On the raw scale the gap is +1.35 [+1.16, +1.55] hundredths of cosine distance, 0.22 of a within-cell standard deviation (0.062): modest but precise. Table 20 gives every specification.

Own work excluded. With every earlier paper by any of the paper's own authors removed from the reference set, newcomer papers still sit +4.33 percentile points [+3.44, +5.22] farther from earlier work than incumbent papers (E); among method papers the gap is +3.71 [+2.70, +4.72], and with team size +4.01 [+3.13, +4.89]. About a quarter of the main gap is therefore the incumbents' closeness to their own earlier work, and the rest is distance from the field's earlier work. The gap remains when the authors' own earlier papers are excluded.

Year	Gap [95% CI]	Cells	New.	Dist.
2019	+1.26 [−5.34, +7.86]	35	86.3	0.473
2020	+5.74 [+2.01, +9.48]	66	81.9	0.443
2021	not interpreted[a]	11	99.5	0.426
2022	+5.14 [+2.03, +8.24]	76	82.1	0.410
2023	+6.22 [+3.93, +8.50]	91	73.5	0.401
2024	+6.28 [+4.49, +8.07]	96	71.2	0.395
2025	+5.04 [+3.50, +6.57]	100	70.1	0.386
2026	+6.40 [+5.13, +7.66]	100	67.2	0.369

Table 21: Gap by year (percentile points). Cells: topic cells with both newcomer and incumbent papers. New.: newcomer share of papers (%). Dist.: mean raw distance from earlier ICLR work. [a] +22.49 [+10.57, +34.40] on 11 incumbent papers. Outcome-free (E).

Contribution type, team size and era. The gap holds within method papers (+5.24 [+4.16, +6.31]) and within method papers other than new architectures (+5.00 [+3.86, +6.14]), so it is not the method-type reversal restated. Team size, class and FM controls barely move it (+5.59; +5.76). It is no smaller in 2023–2026 (+6.01 [+5.03, +6.98]) than in 2019–2022 (+4.95 [+2.32, +7.58]); the difference is +1.06 [−1.64, +3.75] (p 0.44), even though the FM share rose from 10.6% (2023) to 54.4% (2026) within the later era.

By year. The gap is positive in every year from 2020 to 2026, and between +5.0 and +6.4 in every such year except 2021 (Table 21). In 2019 it is +1.26 [−5.34, +7.86] (p 0.70), on only 35 cells with both kinds of paper. The 2021 estimate rests on 11 incumbent papers, a consequence of the identifier switch, and is not interpreted. Raw distance falls every year, from 0.473 (2019) to 0.369 (2026), because a growing field leaves closer earlier neighbours; this is why the comparison ranks papers within topic × year.

Caveats. (1) The analysis was added during writing; it is unregistered, outcome-free and descriptive, and it says nothing about acceptance. (2) Incumbents' own earlier papers can be among their nearest earlier neighbours; the own-work-excluded variant removes them and leaves +4.33. (3) Distance in SPECTER2 space is not originality, and vocabulary from other fields may look distant. (4) Papers are ranked within cells of the all-years map only; the comparison was not re-run on 2017–2020-map cells. (5) The identifier switch turns some incumbents into newcomers after 2020, which pulls the gap towards zero. (6) The measure was recomputed rather than read from the gate module's stored values; a value-by-value comparison with them is not part of the committed output.

What this means for the test. The follow-on prediction expects newcomer papers to sit closer to earlier work than incumbent papers. They sit farther, also when the authors' own earlier papers are excluded. The quarter of the gap that is incumbents' closeness to their own earlier work is itself continuity among producers. A method paper is a contribution type, not a measure of novelty, and this distance is not one either; but neither measure shows newcomers following.

F Other Gate Modules

Section 6 asks whether acceptance goes with following the crowd or with a record. The analyses below are discovery only (D) unless stated. Each was registered as exploratory or declared before it was run; all are associations within topic and year, none is advice, and none changes the reading in Section 6. The planned exploratory analyses are labelled "planned" to separate them from the post hoc far-jump tests E1–E4.

The newcomer gap is mostly about who writes. Newcomer papers are accepted less often within topic × year: −10.2 pp [−11.8, −8.7] with class and FM controls (−10.5 in years with consistent identifiers; planned E2; D). Among 31,549 decided 2019–2026 papers, newcomer papers were accepted at 37.2% and incumbent papers at 47.3%. With 160,222 paper-author rows (81,013 authors, 15,579 of whom wrote both kinds of paper), the gap on the same 28,370 papers is −5.4 [−7.1, −3.7] without and −3.1 [−4.5, −1.6] with author fixed effects; the same author entering a topic new to them is −1.4 pp [−2.7, −0.0] (p 0.049; D). First-author and last-author fixed effects give −2.2 [−4.8, +0.3] and −2.9 [−4.6, −1.1]. ICLR is double-blind, and the residual need not reflect how reviewers treat the person: less topic familiarity can make the paper weaker.

What goes with acceptance among newcomer papers. Against newcomer method papers, newcomer theory papers are +10.2 [+7.2, +13.2] and newcomer FM papers +4.1 [+2.0, +6.2]; dataset/benchmark +2.4, evaluation −0.5 and empirical −3.1 are not significant, and other papers are −9.1. Against teams of 3–4 authors, teams of 1–2 are −8.5, 5–6 +3.7 and 7 or more +11.1. Newcomer papers in more crowded topics are −2.8 per

log unit (Holm 0.02); in hot topics +1.2 (n.s.). The newcomer gap is present in every subgroup, from −5.1 (theory) and −6.4 (teams of 7+) to −14.6 (dataset/benchmark) and −11.6 (most crowded topics), and is negative in all 20 broad areas (−3.8 to −20.8), none surviving Holm correction (D). Selection may account for these: newcomer theory papers may come from established researchers in other fields, and large teams bring seniority.

Team size. Acceptance rises with team size: 28.7% (1–2 authors; 4,176 papers), 37.6% (3–4; 12,022), 41.6% (5–6; 8,843) and 49.0% (7+; 7,149). Within topic × year the gradient is +8.8 pp per doubling [+8.0, +9.7] (+7.8 with a newcomer control; D). It is +13.5 for dataset/benchmark and evaluation papers, +9.3 for method, +5.5 for empirical and +3.3 (n.s.) for theory; equal slopes are not rejected (calibrated p 0.13). Areas differ (calibrated p 0.035), but the slope is positive in every area. Team size stands in for compute, data, engineering and seniority.

Topic insiders. Population: 24,601 decided 2023–2026 papers (acceptance 41.1%), of which 16,579 have no topic insider, 3,555 one, 2,053 two and 2,414 three or more. Besides the supported H1 (Table 13), the module's family gives: H2, concavity, +2.95 (Holm 0.55); H3, the insider contrast in hot minus other topics, −0.8 [−4.9, +3.2]; H4, a proven outsider minus an unproven insider, +4.6 (Holm 0.088); H6, last- minus first-author insider, −2.5 (Holm 0.55); H5 is not reported. Two or more insiders against one: +0.5; familiarity without an earlier acceptance (767 papers) gives +0.2 [−3.8, +4.2]. Robustness of H1: the logit is consistent; an early-map insider gives +3.0 (p 0.003); without earlier versions +3.4; a keyword-defined insider +0.0, a null the registration anticipated because the two definitions correlate only 0.18–0.29. A per-head decomposition (one specification outside the family; D) gives +1.47 [+0.95, +1.99] per author new to ICLR, +4.77 [+4.23, +5.31] per experienced author from other topics and +5.34 [+4.65, +6.04] per topic insider; experienced outsider minus novice +3.30 (p 2.8×10^{-16}), insider minus experienced outsider +0.58 (p 0.16). Per head, an insider is worth about as much as an experienced author from another topic, so the insider association is compatible with general experience and visible status as well as with topic continuity.

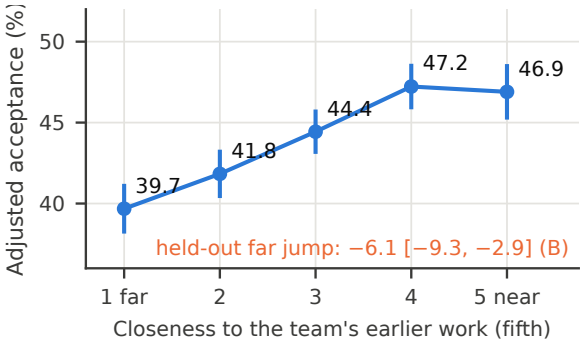


Figure 6: Adjusted acceptance by fifth of closeness to the team's earlier ICLR work ($F_{\max}$), far to near, with 95% intervals: discovery, decided 2023–2026 papers with an experienced author (20,430 papers; shape descriptive, D). Annotation: the declared held-out far-jump test (B; held-out supported in the main text).

Far jump: shape and robustness. Among 20,430 decided 2023–2026 papers with at least one experienced author (acceptance 44.1%), adjusted acceptance by fifth of $F_{\max}$, far to near, is 39.7 [38.1, 41.2], 41.8 [40.3, 43.3], 44.4 [43.1, 45.8], 47.2 [45.8, 48.6] and 46.9% [45.2, 48.6] (3,855 to 4,265 papers per fifth; D shape; Figure 6). Acceptance is lower the farther a paper sits from the record, and the closest fifth is no higher than the fourth. E1 holds with a newcomer control (−5.0), with at least three prior authors (−4.1), without earlier versions (−5.5), with withdrawals after review counted as rejections (−4.9), on the early map (−5.2), with closeness to the last author's record (−4.6) or the whole team's (−4.7), and on link-removed text (−5.39). $F_{\max}$ correlates −0.57 with the newcomer flag, so E1 partly overlaps the newcomer gap. E2, near against median closeness, is +1.75 [−0.17, +3.66] (Holm 0.148) in discovery; by the quoting rule it comes with the held-out +2.34 [−1.01, +5.70] (p 0.17, unregistered) and the pooled +2.10 (Holm 0.036), so "no further gain near the record" is not established. E3, the far-jump contrast in hot minus other topics, is −0.35 [−3.13, +2.42]. Within-author closeness (E4) compares one author's papers with different co-authors. Controls for newcomer status and for co-authors' experience move it little (+0.48 and +0.50 per 0.1 cosine), but an unobserved co-author quality confounder could also bias it: in a simulation reported in its declaration, a planted between-author quality confounder that runs through co-authors moved the estimate. Far means far from the visible ICLR record since 2021; co-author choice and pivots that follow failures are not controlled.

Field distance and twins. The field-distance population is 31,549 decided 2019–2026 papers (acceptance 40.1%). P1b is negative in 11 of 11 variants and Holm-significant in 8 (within class × year −2.0, Holm 0.65; prior-density control −2.4, Holm 0.12; fixed reference −2.5, Holm 0.050); link-removed text gives −3.51. It did not replicate (Table 13). P1a, the most-distant fifth against the middle, is +0.44 [−1.41, +2.28], so maximal distance carries no detectable cost at quintile resolution. A paper has a twin if an independent same-year submission lies at cosine ≥ 0.70; 37.7% of 42,326 papers have one (36,863 independent pairs at 0.70–0.80 and 1,400 at ≥ 0.80, 2018–2026), and the rank correlation between distance and twin count runs from −0.47 (2019) to −0.71 (2026) (E). By the P2 quoting rule, the discovery estimate for having a twin, +0.68 [−0.84, +2.20] (Holm 1.0), comes with the held-out +3.69 [+0.90, +6.48] (p 0.010, unregistered) and the pooled +1.86 [+0.47, +3.24] (Holm 0.063); the held-out estimate does not fit a "no detectable gap" reading. Identical ideas (cosine ≥ 0.80; 692 decided pairs) allow only an underpowered shared-fate test (p 0.15).

Interactions with topic heat. No gate or entry association detectably differs in hot topics. Newcomer × hot: method −0.4, measurement −0.6, acceptance +0.2 pp, all n.s. (planned E5). The insider contrast in hot minus other topics is −0.8 [−4.9, +3.2], the far-jump contrast −0.35 [−3.13, +2.42] (held-out −3.09, unregistered), and the twin association −1.49 [−4.51, +1.52] (held-out +1.47, unregistered). By era, the crowding slope is −0.3, −6.5 and −2.9 pp per log unit (2018–2020, 2021–2023, 2024–2026; all n.s.).

FM premium and new architectures against the tide. FM papers are accepted more often within topic × year: +4.62 [+2.79, +6.44] with class controls (+4.06 without; planned E1; D). Among 24,596 decided 2023–2026 papers (9,908 FM papers), the premium does not detectably change with the topic's prior adoption: +0.19 [−0.96, +1.34] per logit unit (p 0.74). In topics with at least 50% prior adoption, the premium is +2.5 [−1.3, +6.3] (Holm 0.38) on the all-years map and +6.0 [+3.1, +8.8] on the early map. New-architecture papers are accepted relatively more often where prior FM adoption is high: +8.3 [+1.3, +15.4] for 0→100% (p 0.02; planned E8; D). With H1d, one possible reading is scarcity; this is a hypothesis, not a test. A fad premium would be expected to fade with adoption; we detect no such fading, which fits rewarded imitation and efficient choice alike.

Acceptance by contribution class. Against method papers within topic × year (planned E10; D): theory +8.9 [+6.5, +11.3], dataset/benchmark +3.9 [+1.1, +6.7], empirical −2.7 [−5.2, −0.3], evaluation −1.9 (n.s.), other −8.0 (n 32,159).

Withdrawal after review. Withdrawal after review rises with crowding, +1.62 pp per log unit [+0.42, +2.81] (p 0.009; 42,214 papers; planned E3; D); withdrawal before review does not (−0.07, n.s.). Counting these withdrawals as rejections moves H3a to −2.12 (unadjusted; outside the verdict).

Areas and federated learning. Acceptance edges of 68 frequent keywords and 100 topics spread about ±10 pp, but none passes Benjamini–Hochberg correction (minimum q 0.185 for keyword levels), and the declared tests were conservative (permutation rejection 0.5–1.4% at nominal 5%). Federated learning was the starkest discovery case: 475 keyword papers, −11.3 [−17.9, −4.8] (p 0.005, q 0.19); topic-defined, 485 papers, −13.5 [−21.1, −5.9] (q 0.17); adjusted −10.5; trend −3.0 pp per year [−6.4, +0.5]. It was not confirmed on held-out papers (Table 13). Three mechanism probes were declared. Composition and influx: not supported; the share of papers with an author with an earlier accepted ICLR paper rose from 4% (2021) to about 60% (2023–2025), keyword newcomers fell from 96% to 40–58%, team size (4.3–4.9) and the method share (about 85%) were stable, and composition adjustment keeps 76% of the raw trend (adjusted −2.2 pp per year, p 0.20). The field moved on: supported descriptively, with the keyword's share peaking at 3.2% (2023) and falling to 1.1% (2026). Harsher judgement: consistent with the data, not testable here; the decline appears in every author group (none significant), and the registered reading rule returns "mixed".

Keyword timing. The A1 population is 22,302 decided 2019–2026 papers (40,074 keyword rows); the Mainstream acceptance rate is 37.8%. A1 is positive in 7 of 7 variants (+2.3 to +4.1, and the logit). Cooling minus Mainstream is +0.13 (Holm 0.90), and the age gradient +1.74 (Holm 0.077), not supported. Old rising keywords give +3.9 nominally, so A1 may reflect a low Mainstream baseline as much as keyword age; within a keyword, rising

against flat years give +0.9 [−0.8, +2.7]. Age, size and breadth of keywords are confounded. In PH6, relative to each wave's own level, the discovery gap is +9 to +30 pp 2–5 years before the peak (on few papers), near 0 around the peak and −2 to −5 pp 3–5 years after it: the decline starts before the peak.

Topic edges and the keyword winner's curse. The noise-free SD of topic acceptance edges is 6.4 pp [5.0, 7.2] on the early map (6.1 on the all-years map; keywords 4.7; areas 4.4); a single year's rate is half noise at about 58 decided papers. The registered unit for persistence is the early-map topic; keyword units show no detectable persistence on held-out papers (b +0.16 [−0.34, +0.66], unregistered). The empirical-Bayes tie rule was changed after outcome-free outputs were seen (disclosed). On the producer side (outcome-free; post hoc; D/E), the keywords with the largest raw growth in one year are among the top growers the next year less often than the base rate (16.0% against 20.6%; lift −0.045 [−0.075, −0.017]; Holm 0.018; 2,491 keyword-years), while those with the largest shrunk growth persist (40.3% against 22.1%; Holm 0.006). The raw reversal is confined to a one-year horizon and to keywords with 10–19 users, and on topic maps the raw lift is not negative (early map +0.04, Holm 0.25; all-years map +0.11). Last year's fastest-growing keywords are thus partly noise, as topics' past acceptance edges are.

Link-removed text. URLs and availability sentences were removed from the abstracts of decided 2019–2026 discovery papers, which changed the text of 22.4% of accepted and 8.5% of rejected papers, and the abstracts were re-embedded. The declared far-jump and field-distance families were rerun with frozen code; their robustness variants were not: E1 −5.44 → −5.39 (Holm 2.0×10^{-8}), E4 +0.55 → +0.53 (Holm 0.009), P1b −3.27 → −3.51 (Holm 0.0004), P7 +0.39 → +0.46 (Holm 0.035; consistent, not upgraded), P2 +0.68 → +0.98 (n.s.). The check does not cover rewritten results or polished wording, labels, topics or titles, and no link-removed text exists for held-out papers.

What this means for the test. None of these modules shows a premium that changes with how many others follow. Interactions with topic heat are null, and the edges of both topics and keywords regress. The FM premium does not detectably change with adoption, which fits rewarded imitation and efficient choice alike (Section 2); at the package level the gate cannot separate the routes. What acceptance does go with, beyond the record, is who writes (team size, experience), which is why Section 6 leads with the far jump among papers that all have an experienced author and with the within-author slope, which hold more of the writer fixed.

G Title Style

Surface form is where an imitation-like spread is plainest. This appendix gives the definitions, the yearly shares and the declared acceptance test behind Section 6. All numbers come from the committed outputs of the PH7 module (`research/fashion/output_pooled/posthoc/ph5_titles/`: `shares.csv`, `tests.csv`, `rates.csv`).

Definitions. The prefix is the text before a title's first colon, and the subtitle after it must be nonempty. D1 (broad): the prefix has 1–3 whitespace tokens. D2 (name-like; the primary definition): D1, every prefix token starts with an upper-case letter or a digit, and the first token is not a generic lead word (for example Towards, Beyond, Rethinking, Learning, Understanding, Enhancing, On, How, What, Position). D3 (coined single token): one token with at least two capitals, or a digit and a capital (as in BERT, LoRA, GraphSAGE, Llama-2). D4 (one name-like token): D2 restricted to exactly one token. Shares are computed over the 43,480-paper discovery corpus (Table 5); no outcome column is read for them.

Shares. Under every definition the share of titles that open with a short coined name and a colon rose roughly fourfold, from 7–14% (2017) to 37–49% (2026), by 3.4 times (D1) to 5.7 times (D3), and it rises every year after 2021 (Table 22). The level depends on the definition by about ±4 pp in 2017 and ±6 pp in 2026; the trend does not. The 2017 share rests on 384 titles (sampling SE about 1.6 pp at 11%).

The declared acceptance test (PH7). The test was declared in the pre-registration log before it was run (2026-11-04) and run once; discovery and held-out estimates are computed separately. The population is the registered acceptance population (decided ordinary submissions, 2018–2026: 32,190 discovery and 8,185 held-out papers). The model is a linear probability model of acceptance on the

Year	n	D1	D2	D3	D4
2017	384	14.32	10.94	6.51	7.55
2018	770	16.88	13.90	7.92	8.83
2019	1,248	19.95	17.71	10.74	11.86
2020	2,070	21.40	18.79	13.77	14.64
2021	2,360	21.10	18.77	13.18	14.11
2022	2,649	24.61	21.93	15.97	17.25
2023	3,872	25.88	23.22	17.61	18.96
2024	5,765	32.91	30.20	23.94	25.26
2025	8,951	40.74	37.56	29.33	31.25
2026	15,411	48.83	44.57	37.00	38.76

Table 22: Share of discovery submissions (%) whose title is branded under definitions D1–D4, by year. Record: `shares.csv` (discovery rows). Outcome-free (E).

branded-title indicator with topic × year fixed effects and class, FM and $\log_2$ team-size controls, with errors clustered by topic; the raw model has year fixed effects only. The declared rule: the held-out D2 estimate is supported if it is positive with one-sided $p < 0.05$; otherwise "no premium of 2 pp or more" if its 90% interval lies inside ± 2 pp; otherwise not supported, and uninformative if it lies below its MDE80. Table 23 gives every estimate.

Result. The declared held-out test gives +1.76 [−0.82, +4.34], one-sided p 0.089, MDE80 3.23: not supported, and uninformative. Its 90% interval [−0.40, +3.92] does not lie inside ± 2 pp, so "no premium of 2 pp or more" cannot be claimed either. In discovery, branded titles are accepted +1.74 [+0.37, +3.11] more often, and all four definitions give intervals above 0 (+1.58 to +1.78), but these are not tests. Raw acceptance is 42.5% for branded and 38.9% for other titles in discovery (10,395 and 21,795 papers), and 42.4% and 37.9% held-out (2,685 and 5,500). The raw within-year gaps (+3.31 and +3.97) are about twice the controlled ones, so about half of the raw gap goes with topic, class, FM use and team size. Whether the gap changed in 2023–2026 cannot be told (−0.81 in discovery, −5.04 held-out, both intervals including 0). The test supersedes an earlier exploratory estimate whose code was not kept: on the same papers, branded titles are accepted somewhat more often, not at the submitted rate.

What this means for the test. The title form spread roughly fourfold, the plainest imitation-like spread in the corpus. It goes with a small acceptance premium in discovery, which is not a test, and the declared held-out estimate has the same size but is not supported. Titles can be revised after the decision (camera-ready), which could create such a premium, and this cannot be checked here. Whether the spread is a bandwagon that goes unrewarded (Abrahamson and Rosenkopf, 1993) or carries a small premium stays open; the main text labels it exploratory.

Split and model	D2 (primary)	D1	D3	D4
Discovery, controls (not a test)	+1.74 [+0.37, +3.11]	+1.78 [+0.42, +3.14]	+1.60 [+0.19, +3.02]	+1.58 [+0.18, +2.97]
Discovery, raw (year FE only)	+3.31 [+1.75, +4.87]	+3.26 [+1.78, +4.75]	+3.34 [+1.79, +4.89]	+3.39 [+1.82, +4.96]
Discovery, branded × 2023–2026	−0.81 [−3.89, +2.27]	−1.49 [−4.67, +1.68]	−1.74 [−5.13, +1.66]	−0.69 [−3.86, +2.49]
Held-out, controls (declared test for D2)	+1.76 [−0.82, +4.34]	+2.08 [−0.52, +4.67]	+0.62 [−1.90, +3.13]	+1.50 [−1.10, +4.09]
one-sided p; MDE80	0.089; 3.23	0.058; 3.25	0.313; 3.15	0.128; 3.25
Held-out, raw (year FE only)	+3.97 [+1.32, +6.62]	+4.05 [+1.37, +6.73]	+3.55 [+1.03, +6.06]	+4.15 [+1.54, +6.77]
Held-out, branded × 2023–2026	−5.04 [−11.53, +1.44]	−4.78 [−10.79, +1.23]	−0.56 [−8.54, +7.41]	−1.58 [−9.40, +6.25]

Table 23: PH7: acceptance of branded against other titles, pp with 95% intervals (discovery 32,190 papers, held-out 8,185). Controls: topic × year fixed effects, class, FM and $\log_2$ team size. Raw: year fixed effects only. Branded × 2023–2026: the change in the gap in the late years. MDE80 computed in the declared run. Title style had been explored before registration, so no discovery row is a test. Record: `tests.csv`.

H Size Calibration

A registered test is only as good as its size and power. This appendix reports both, including a size failure in the entry tests on which the reversal of the herding prediction rests.

Dry runs. Before the freeze, two dry runs of 12,400 simulated jobs each (same seeds) measured size, false support under the full decision rule, and power. The entry nulls were calibrated on the corpus's own topic-year structure; the gate nulls used fixed parameters (base rate 0.32, two variance settings) plus a within-year permutation of the real acceptance outcomes; power grids ran from 0.5 to 6 times the smallest effect of interest (SESOI).

The size failure. Under an entry null whose structure follows the 2017–2020 map, and so cuts across the all-years topics used for clustering, the main tests of H2b and H2c rejected 12.6% and 10.0% of the time at nominal 5%; all other sizes were 3.8–7.2%. As registered, their confirmatory p-values came from a wild cluster restricted bootstrap (Rademacher weights by topic, null imposed, 9,999 draws; Cameron et al., 2008). The validation run showed that the bootstrap does not fix the failure (12.4% and 9.6%); we report it rather than hide it. Two-way clustering by all-years and early-map topic (S11) had sizes of 4.4–7.6% in every null (7.6% and 7.0% in the cross-map null). Under the cross-map null, some entry hypothesis was supported in 6.4% of replications (2.0% under the all-years null), some directional claim was made in 11.2% (4.6%), and per-hypothesis false support was at most 3.2%. The registered rule offers no extra protection against such structure: all main-specification false positives also passed the robustness and hindsight requirements.

Why the entry reversal survives it. The reversed estimates are large relative to the simulated inflation: $t = 6.7$ and -4.4, against t-inflation factors of 1.31 (H2b) and 1.22 (H2c) in the one cross-map scenario. Dividing by these factors still gives Holm p of 4×10^{-6} and 1×10^{-3}, and S11 agrees (Holm p 4×10^{-8} and 2×10^{-4}). The factors are a pre-declared single-scenario bracket, not a correction; how strong cross-topic structure is in the real data is unknown, so these p-values and their intervals remain approximate. On held-out papers, the simulated one-sided null sizes were 4.5% (H1d), 2.0% (H2b′), 2.5% (H2c′), 5.0% (H3a) and 4.0% (H4-FL).

Power and reading rules. Table 24 collects the minimum detectable effects. Non-supported tests below their MDE80 are uninformative, not evidence of no association. The gate's crowding test therefore cannot tell effects below about 4–5 pp per log unit from zero, so "no detectable premium" is the strongest statement it supports. Supported held-out tests whose MDE80 exceeds the discovery estimate confirm direction, not size. The federated-learning level test could easily miss a gap of the discovery size (MDE80 14.5 against −11.3). The post hoc area tests were conservative (permutation rejection 0.5–1.4% at nominal 5%).

What this means for the test. The size failure is disclosed, not corrected, and it bears on the entry tests; their reversal is large relative to it and held on held-out papers, whose simulated sizes were at or below 5%. The absence of a detectable crowd

Test	SESOI	Disc.	Held-out	Est. / power
H1d	–	–	12.0	−7.98
H2a	2	3.6	–	−0.73
H2b / H2b′	2	1.7	2.8	+3.71
H2c / H2c′	1	1.1	2.4	−1.70
H3a	1	4.1 (2.8)	4.8	54% (88%)[p]
H3b	1	2.4 (2.1)	–	−0.36
H4-FL level	–	–	14.5	−11.3
E1 far jump	–	–	4.9	power 0.89
P1b	–	–	5.4	power 0.47
P7	–	–	≈1.0–1.1[d]	+0.39
Teams H1	–	–	≈4.5[d]	+3.42
A1	–	–	≈4.3[d]	+2.67
PH6 D	–	3.6	7.4	−5.7
PH7 titles (D2)	–	–	3.2[r]	+1.74

Table 24: Smallest effects of interest (SESOI, ±) and MDE80 in pp (H3a, H3b per log unit). Disc.: frozen discovery dry run (high-variance setting; low-variance in parentheses). Held-out: registered, simulated at each test's Holm first step; [d] derived post hoc ($2.49\times$SE). PH6: discovery MDE80 as reported with its discovery estimate, which is not a test. Est.: discovery estimate, for comparison; power at the discovery effect for E1 and P1b. [p] H3a power at three times the SESOI in the discovery dry run. [r] computed analytically in the declared PH7 run.

premium is a statement about effects above roughly 5 pp per log unit, not about smaller ones.